

Pathophysiology of Proteinuria

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Norberto Perico | Andrea Remuzzi | Giuseppe Remuzzi

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MECHANISMS OF PROTEINURIA

One of the most common features of glomerular diseases is an abnormal excretion of plasma proteins in the urine. Complex changes in the structure and function of the glomerular capillary, as well as the entire nephron, are responsible for the final elevation in urine protein concentration in several kidney disorders. In this chapter, we review the pathophysiology of urinary protein excretion.

Functional properties of the glomerular filtration barrier, tubular interaction with filtered proteins, and characterization of structural molecules relevant to the filtration barrier in glomerular endothelial cells and in particular on podocytes have been reported.¹⁻³ There are two distinct phenomena that can result in proteinuria. The first is elevation of glomerular filtration of plasma proteins that are almost completely retained in the circulating plasma in normal physiologic conditions; the second is an incomplete reabsorption of proteins by the proximal tubule. The two phenomena are interrelated and likely are both present in proteinuria when proteins the size of albumin and larger are present in urine. Despite several experimental and clinical observations,⁴ the precise nature of the functional changes responsible and their quantification remain the subject of numerous ongoing investigations.⁵

STRUCTURE AND FUNCTION OF THE GLOMERULAR CAPILLARY WALL

The glomerular capillary allows filtration of a large amount of water and small solutes while efficiently restricting protein macromolecules within blood circulation. This selective function is specific to the glomerular capillary membrane, which is far more permeable to water than any other capillary membrane. With the development of glomerular diseases, the changes in capillary membrane structure at molecular and/or cellular level result in loss of hydraulic permeability, reduction in filtering surface area, and loss of glomerular filtration rate (GFR). Despite the reduction

in permeability to water, the capillary membrane becomes more permeable to circulating macromolecules.

GLOMERULAR CAPILLARY WALL ORGANIZATION

Studies available in the literature describe in detail the complex organization of the glomerular capillary (see [Chapter 2](#)). Thus while the glomerular capillary comprises numerous branching segments, the glomerular capillary organization has usually been considered as a set of several uniform segments in parallel. Similarly, the capillary membrane has been considered as a uniform three-layer structure. Recent investigations allow a better understanding of the functional effects of geometrical and spatial organization of the glomerular capillary. These aspects have revealed some new insights in the mechanisms responsible for glomerular capillary dysfunction.

Glomerular capillary network

According to classical optical microscopy, the capillary network is composed of numerous capillary segments connecting afferent and efferent arteriole within a tuft that has a mean diameter of 120 to 150 μm . Specific investigations with reconstructions from serial sections,⁶ or confocal microscopy,⁷ allow investigation of capillary segment organization and, in particular, calculation of blood flow distribution and water filtration along the network.⁸ Due to the large number of capillary segments and their apparent size, blood flow is expected to be uniformly distributed along the network with low blood velocity. This allows the blood to remain in close contact with the filtration membrane. However, more detailed geometric reconstructions show that the network has some heterogeneity. The size of some capillary segments (with diameter $<3\text{--}4\ \mu\text{m}$)⁶ would suggest that they are perfused only by plasma and may represent a sort of shunt in the network to decrease overall network pressure. This finely organized geometry is the result of cellular organization and remodeling and is importantly affected by disease processes resulting in simplification of capillary network, ultimately resulting in capillary obliteration in areas of segmental sclerosis.⁹ These local hemodynamic changes affect the filtration function of the

capillary network, as elevation of blood flow and pressure is expected to occur in some capillary segments leading to abnormal filtration of circulating proteins.¹⁰

Glomerular capillary wall

At a smaller scale, the organization of glomerular membrane is rather heterogeneous. The arrangement of the capillary wall that is considered the filtering surface is characterized by the three-layer composition consisting of endothelial cells, glomerular basement membrane (GBM), and epithelial cells.

The structure of the epithelial cells also affects water and macromolecule filtration. As reported by Neal and coworkers,¹¹ a large fraction of the filtration membrane is covered by epithelial cell bodies or the presence of adjacent epithelial cells. The three-dimensional spaces created by these structures are called “subpodocyte space (SPS)” and “interpodocyte space (IPS),” respectively. Theoretical analysis of the transport of both water and macromolecules through the SPS¹² indicates that structural organization of this compartment induces significant resistance to water filtration from glomerular membrane to the urinary space. Macromolecule transport may also be influenced by the SPS.^{13,14}

ULTRASTRUCTURE OF THE GLOMERULAR CAPILLARY MEMBRANE

Endothelial cell layer

Glomerular endothelial cells are the most fenestrated in the circulation, with a pore area in peripheral zone that occupies from 20% to 50% of the cell surface.¹⁵ The surface of endothelial cells has been considered to have a negative electric charge due to the presence of electrical charges of glycoproteins, glycosaminoglycans, and membrane-associated proteoglycans (glycocalyx).¹⁶ These negative charges act as an electrostatic barrier to the transmural passage of anionic circulating proteins, such as albumin. Thus even if endothelial fenestrae are much larger than albumin (about 60 nm in diameter as compared with a radius of 3.6 nm for albumin), negatively charged circulating macromolecules remain within the circulation. It has been demonstrated¹⁷ that disruption of the endothelial glycocalyx increases glomerular albumin filtration even in the presence of only minor changes in both the GBM and glomerular epithelial cells. The role of the glomerular polysaccharide-rich endothelial surface layer (ESL) to act as a filtration barrier for large molecules such as albumin has been recently confirmed in C57Bl/6 mice given long-term infusion of hyaluronidase, a hyaluronan degradation enzyme that disrupts the endothelial glycocalyx proteoglycans.^{18,19} A new electron microscopy technique that allows visualization of the ESL and albumin transport within the entire glomerular section at nanometer resolution was used in this set of experiments.²⁰ It was shown that glomerular fenestrae are filled with dense negatively charged polysaccharide structures that are largely removed in the presence of circulating hyaluronidase, leaving the polysaccharide surface of other glomerular cells intact.¹⁸ Hyaluronidase treatment, however, induced albumin passage across the endothelium in 90% of glomeruli, whereas this could not be observed in untreated control animals. Thus the

polyanionic ESL component hyaluronan is a key component of the glomerular endothelial permeability barrier whose reduction facilitates albumin passage across the endothelial layer and the capillary membrane.

Glomerular basement membrane organization

The basement membrane layer that characterizes the capillary wall (see [Chapter 2](#)) has been shown to exert an important contribution to protein retention by the capillary wall. The molecular composition and organization of this basement membrane suggest a sieving function due to both macromolecule size and charge.²¹ Structural proteins such as collagen type IV and laminin, as well as heparan sulfate proteoglycans, represent not only a steric hindrance but also a charge effect on the filtration of plasma proteins.²² Thus changes in composition and/or organization of GBM molecules are expected to reduce water filtration and retention of circulating macromolecules.²³ Mutations in the *COL4A3*, *COL4A4*, or *COL4A5* genes that encode collagen type IV $\alpha 3$, $\alpha 4$, and $\alpha 5$ chains, respectively, cause Alport syndrome, a hereditary glomerular, auditory, and ocular disease.²⁴ Mutations in the gene encoding laminin $\beta 2$ (*LAMB2*) cause Pierson syndrome, a congenital nephrotic syndrome with associated extrarenal manifestation.²⁵ Studies using knockout mouse models of Alport and Pierson syndromes have documented that GBM, defective of these specific components, is more permeable to ferritin or albumin than is the normal GBM, indicating it has a role in glomerular permselectivity.^{26,27}

Tight adherence to the basement membrane is required to prevent podocyte detachment into Bowman capsule. On a molecular level, a multitude of adhesion receptors including heterodimeric integrins mediate interaction of cells with the surrounding basement membrane.^{28,29} One common form of integrin-mediated adhesion is focal adhesions (FAs), which have been extensively studied in cultured cells.²⁹ Recently, FERM-domain protein EPB41L5 has been identified as a highly enriched podocyte-specific FA component.³⁰ This provided the clue to document that genetic deletion of the related *Epb41l5* gene resulted in severe proteinuria, detachment of podocytes, and development of focal and segmental glomerulosclerosis.³⁰

Epithelial filtration slits

A large amount of experimental and clinical research has been generated in the past few decades on the molecular and structural composition of the epithelial junctional complex, known as “filtration slits.” The characterization of several molecular components of this structure³¹ has allowed detailed definition of the proteins that compose the filtration slits ([Fig. 29.1](#)). However, detailed information on the ultrastructure of this intracellular junction is still under investigation.⁵ The original observations by Rodewald and Karnowsky³² suggested a zipper-like structure of the epithelial filtration slit, with rectangular openings of 4 by 14 nm. These dimensions are in contrast with the observation that a limited amount of albumin can traverse the filtration barrier in physiological conditions¹, since the mean molecular radius of albumin is 3.6 nm. Observations with high-resolution scanning electron microscopy (SEM) and 3D electron microscopy reconstruction³³ suggest that the filtration slits are perforated by larger openings of the size

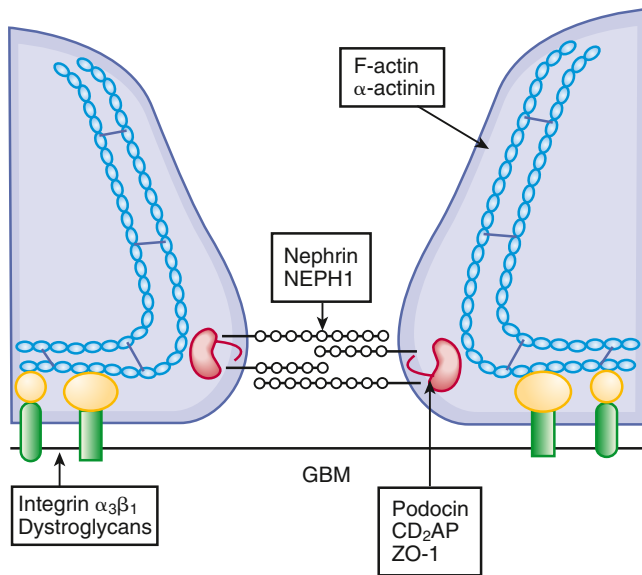


Fig. 29.1 Hypothetical model of the podocyte slit diaphragm. See text for discussion. (Reproduced from Jalanko H. Pathogenesis of proteinuria: lessons learned from nephrin and podocin. *Pediatr Nephrol.* 2003;18:487–491, with permission.)

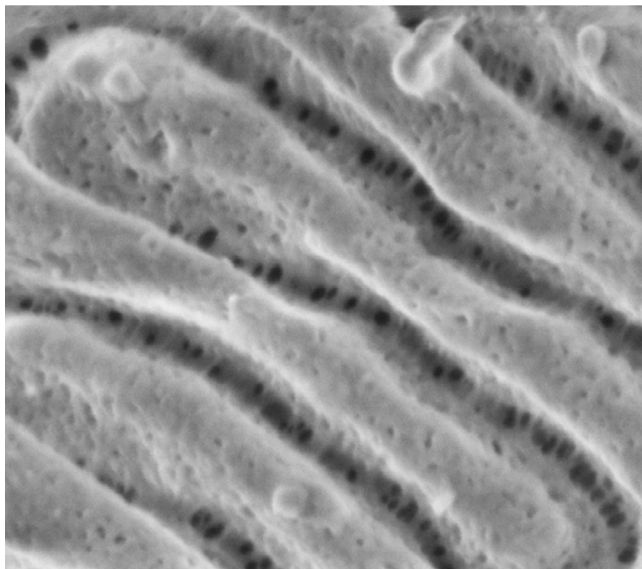


Fig. 29.2 Visualization of epithelial filtration slits obtained using SEM and an in-lens detector to enhance electron detection. Sample was obtained from a Wistar rat, dehydrated with a critical point dryer. The ultrastructure of the filtration slit appears different from the model conventionally proposed in the literature. The radius of the circular pores averages 12 nm. (Reproduced from Gagliardini E, Conti S, Benigni A, et al. Imaging of the porous ultrastructure of the glomerular epithelial filtration slit. *J Am Soc Nephrol.* 2010;21:2081–2089.)

of albumin, with more complex geometry. The morphology of the filtration slit has been further imaged with high-resolution SEM, providing evidence of a new ultrastructure composed of circular pores of different sizes with an average radius of 12 nm^{34,35} (Fig. 29.2). Despite the small size of filtration slit openings, a large amount of plasma water is filtered because of the high-filtration slit length per unit surface area. As mentioned earlier, under physiologic

conditions about 20% of peripheral capillary filtering surface is directly in communication with the Bowman space and epithelial slits are the last resistance encountered by water and filtered solutes across glomerular membrane.¹¹ It should be considered that the flow of ultrafiltrate over the cell surface directly causes shear stress (SS) on the podocyte membrane associated with the filtration slit, as well as to the cell body. In animal models of solitary kidney where filtrate flow is increased, SS on podocyte surface increases 1.5- to 2-fold.³⁶ These forces acting parallel to the GBM are balanced by the mechanical resistance produced by the slit-diaphragm complex, opposing foot processes, preventing widening of the slit.³⁷ On the other side, SS acting perpendicular to the GBM plane tends to detach the foot process from the GBM. However, these shear forces are balanced by transmission of this mechanical load into the cytoskeleton apparatus. These complex mechanical challenges make mechanotransduction an important function of podocytes.

THEORETICAL MODELS OF GLOMERULAR PERMSELECTIVITY

In addition to structural investigation, functional evaluations of the glomerular capillary wall permselectivity have been extensively used to characterize physiologic conditions and quantify the effects of pathologic changes. These studies are based on the estimation of filtration of endogenous plasma molecules, such as albumin, IgG, and other proteins, or on the use of test macromolecules of different sizes, either neutral or electrically charged. Macromolecule filtration depends on convective and diffusive transport, which are influenced by glomerular hemodynamic conditions and water filtration. As described later, several investigators have developed theoretical models to derive intrinsic sieving properties of the capillary wall from estimation of macromolecule filtration in experimental and human studies.

Heteroporous models of glomerular size selectivity

The most used theoretical models of glomerular size-selective function are based on the assumption of water-filled pores of different sizes, as functional equivalents of the glomerular membrane. The passage of water is calculated along the network taking into account the balance between hydraulic and oncotic pressure, as well as membrane hydraulic permeability.^{9,38} For the calculation of solute filtration, the convective and diffusive transport are taken into account, while pore resistance to solute filtration is based on steric and hydrodynamic hindrance.^{10,38} The use of these models indicated that glomerular hypothetical pores have a mean radius of 4.5 to 5.0 nm in humans and a lognormal statistical distribution of pore size around the mean. However, the best simulation of experimental measurements has been obtained assuming that in parallel to restrictive pores, there is a nonselective shunt pathway.³⁹ Application of these theoretical models clearly showed that in several proteinuric conditions, the increased glomerular filtration of largest neutral test macromolecules is not associated with an important change in the size of restrictive pores but rather changes in the nonselective shunt pathway.⁴⁰ This suggests that in normal conditions the small amount of albumin present in the urine may be the result of a small amount of protein filtration that takes place in some

focal areas of the epithelial junction, while the majority of the filtration slits retain the protein.

Fiber models of glomerular size selectivity

Fiber models of solute filtration across the glomerular membrane have also been developed and used.¹ Similar to porous models, the fiber models allow separation of the effect of glomerular hemodynamic changes from those related to intrinsic changes of glomerular membrane selective properties. The advantage of the fiber model is that besides steric hindrance, the effect of membrane and protein electrical charge can be embedded in the model, allowing estimation of changes in membrane properties in terms of both molecular structural organization and electric charge.⁴¹ This modeling approach indicates that filtration of albumin is importantly affected by electrical charge, while on the basis of size selectivity alone, the molecule could easily escape the capillary membrane.

Multilayer membrane models

The structural complexity of the glomerular capillary wall suggested a need to develop more complex theoretical models to simulate more reliably the resistance to water and solute movement across the membrane. These models have been developed and tested by Deen and coworkers²¹ with the aim to estimate the role of individual layers on the filtration of water and solutes. In these models, the resistance of endothelial cells, GBM, and epithelial cells is assumed to act in series. In normal conditions, hydraulic resistance of the endothelial layer is negligible, while GBM and epithelial resistance are comparable. The contribution of the three layers to solute hindrance has been considered, and the major contribution to membrane selectivity is exerted by the filtration slit.⁴² While these models describe in detail the physical interaction of water and macromolecules with the membrane structure, their application is difficult because they require extensive measurement of ultrastructural parameters.

Models of glomerular charge selectivity

As mentioned previously, the fact that negative electrical charges are present in the glomerular membrane (in the glycocalyx of endothelial cells, the negatively charged heparan sulfate of the GBM and the glycoproteins of the cell membrane of podocytes) strongly suggests that circulating proteins that are negatively charged, like albumin, are restricted within the circulation for not only their size but also electrical charges. The use of theoretical models for the simulation of charge-selective function of the glomerular membrane allowed estimation of the amount of electrical charge present within the membrane.¹⁰ These studies indicated that electrical charge is an important component of glomerular permselective function, and changes in membrane electrical charge can explain abnormal albumin filtration even without changes in membrane structural parameters such as pore size or fiber size and length per unit volume. However, the use of these models to investigate glomerular membrane charge is limited by difficulties in measuring glomerular filtration of charged test solutes that interfere with circulating macromolecules and are not filtered simply on the basis of their molecular shape and electric charge alone.⁴³

PROTEIN REABSORPTION BY THE PROXIMAL TUBULE

PROXIMAL TUBULE STRUCTURE AND FUNCTION

The glomerular ultrafiltrate, once flowing inside the proximal tubule, undergoes important changes in composition due to processes of water and solute reabsorption. Besides small solutes and electrolytes, proteins such as albumin are also reabsorbed.⁴⁴ Thus final urinary excretion of proteins depends largely on the interaction of proteins with proximal tubular epithelial cells. These cells form a compact epithelial layer with a basal side in contact with a tubular basement membrane, an intercellular junction, and a luminal surface in contact with tubular fluid. They are characterized by many mitochondria, an index of important metabolic activity, and a prominent layer of microvilli.

The microvillar membrane is the site for receptor-mediated endocytosis of low-density lipoprotein and negatively charged proteins. Albumin in the proximal tubule undergoes specific binding to the extracellular domain of a membrane receptor complex, megalin-cubilin receptor⁴⁴ (Fig. 29.3) and then internalization of the protein by membrane vesicles. These vesicles are then processed for protein degradation, amino acid transport to basal membrane of tubule cells, and release into interstitial space and ultimately into peritubular capillaries. Receptors are recycled to the luminal cell membrane by dense apical tubules.

The amount of albumin filtered at the glomerular level and reabsorbed by proximal tubule cells is not easy to quantify. Ideally, one would have to sample the early proximal tubule and quantify albumin concentration in these microsamples. Despite technical difficulties, micropuncture techniques have been used to avoid sample contamination with plasma present near the puncture site (in interstitial space and peritubular capillaries). The protein concentration in the urinary space was estimated to range from 10 to 25 $\mu\text{g}/\text{mL}$.^{45,46} More recently, direct in vivo imaging of fluorescent albumin by two-photon microscopy has been used to directly estimate albumin concentration in the Bowman capsule fluid.⁴⁷ It has been recently demonstrated that albumin concentration is around 60 $\mu\text{g}/\text{mL}$ in the Bowman capsule in normal conditions in the rat,⁴⁸ corresponding to a fractional clearance of 0.002. Once filtered at the glomerular level, albumin and smaller proteins are almost entirely reabsorbed at the proximal tubular level. In pathologic conditions, when the filtered load overwhelms the reabsorptive capacity, proteins are detected in the urine.

THEORETICAL MODELS OF TUBULAR REABSORPTION

The process of albumin reabsorption by proximal tubule cells has been modeled⁴⁹ to allow a quantitative assessment of the relationship among albumin ultrafiltration at glomerular level, proximal tubular uptake, and final excretion in the urine. In this model, the process of diffusion of albumin across the microvillar space and the uptake of the protein by tubule cell receptors are taken into consideration. The amount of albumin that is reabsorbed during the proximal tubule passage is simulated, assuming the presence of a high-affinity site for binding and internalization of albumin at the base of tubule cell microvilli. According to in vitro and ex vivo data, these receptors are assumed to be half-saturated at concentrations

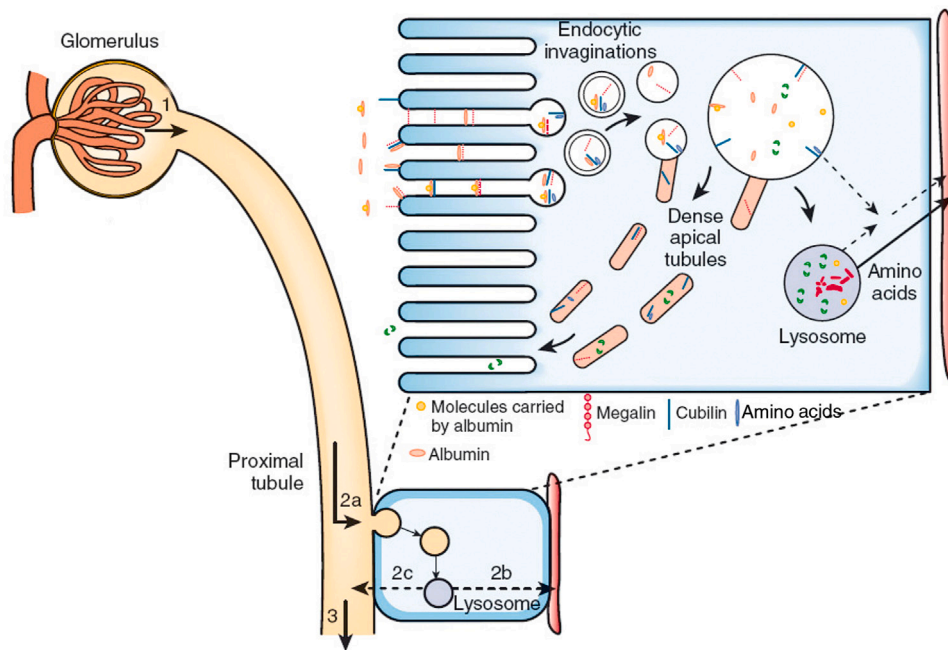


Fig. 29.3 Pathways of albumin degradation in the proximal tubule. Albumin is filtered in the glomeruli (1) and reabsorbed by the proximal tubule cells by receptor-mediated endocytosis (2a). Internalization by endocytosis is followed by transport into lysosomes for degradation. Some intact albumin may escape tubular reabsorption (3), the amount being greater as the glomerular filtration fraction of albumin increases or tubular function is compromised. The upper right shows a schematic representation of the intracellular pathways following endocytic uptake of albumin and possible associated substances. Following binding to the receptors, cubilin or megalin, the receptor-albumin complex is directed into coated pits for endocytosis. The complex dissociates following vesicular acidification, most likely also leading to the release of any bound substances. Albumin is transferred to the lysosomal compartment for degradation. Some albumin may be degraded within a late endocytic compartment and recycled as fragments to be released at the luminal surface. Alternatively, albumin fragments may be recycled from the lysosomal compartment by a yet unknown route. Receptors recycle through dense apical tubules, whereas released substances carried by albumin may be released into the cytosol or transported across the tubular cell. (With permission from Birn H, Christensen EI. Renal albumin absorption in physiology and pathology. *Kidney Int.* 2006;69:440–449.)

like those mentioned for albumin in the Bowman capsule (20–30 $\mu\text{g/mL}$). The effect of the assumption of different values for the maximum absorptive capacity (V_{max}) on the albumin concentration along the proximal tubule is reported in Fig. 29.4. This modeling approach showed that the transport of albumin across the microvillar space has a modest effect on the value of V_{max} needed to fit micropuncture data.

The two most important parameters that determine the fraction of reabsorption of albumin are the single-nephron glomerular filtration rate (SNGFR) and albumin concentration in the filtrate fluid (C_b). A 50% increase in SNGFR is predicted to cause a fourfold to fivefold increase in albumin excretion in rats and humans. For large increases in C_b , such as those measured by micropuncture,⁵⁰ there is a threshold above which the reabsorption of albumin is overwhelmed and the protein appears in the urine. According to theoretical analysis,⁴⁹ this value corresponds to an albumin fractional clearance of approximately 0.001. The combination of increased SNGFR and elevation in filtrate albumin concentration are shown to have important additive effects.

PROTEINURIA OF GLOMERULAR ORIGIN

Changes in glomerular protein filtration and/or defects in tubular reabsorption cause the appearance of proteins in the urine. At values exceeding 300 mg per day, or 200 mg/L , the

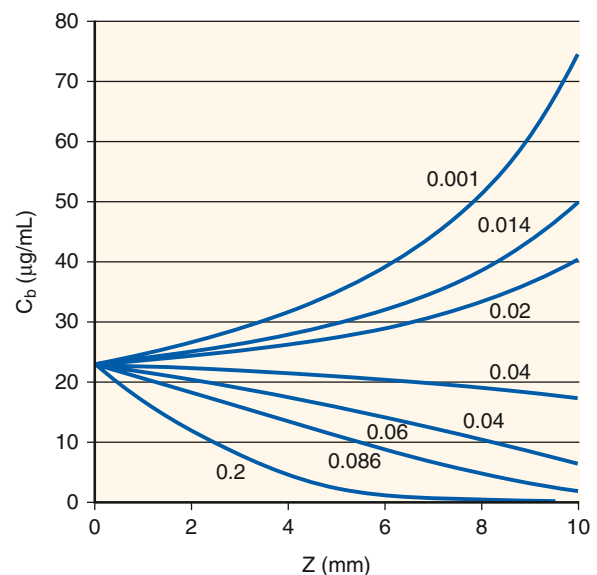


Fig. 29.4 Theoretical calculation of albumin concentration along the proximal tubule. Predictions of bulk albumin concentration (C_b) vs. axial position (z) in rats for V_{max} ranging from 0.001 to 0.2 $\text{ng s}^{-1} \text{mm}^{-2}$. The curve that most closely corresponds to normal rats is that for $V_{\text{max}} = 0.086 \text{ ng s}^{-1} \text{mm}^{-2}$. (With permission from Lazzara MJ, Deen WM. Model of albumin reabsorption in the proximal tubule. *Am J Physiol Renal Physiol.* 2007;292:430–439.)

condition is termed *proteinuria*. Smaller amounts of proteins may appear in the urine in the early stages of progressive diseases, such as diabetic nephropathy. In this case, an albumin excretion between 30 and 300 mg/day (20–200 mg/L) is termed *microalbuminuria*.⁵¹ Proteinuria is considered severe or in the “nephrotic range” when protein excretion is greater than 3.5 g/day.¹ When proteins in the urine have large molecular weight, they are considered to have glomerular origin. If the molecular weight is low, there is evidence that the defect causing proteinuria is likely related to abnormal proximal tubular reabsorption, often related to toxic damage of tubule cells.⁵² Proteinuria associated with progressive kidney disease is predominantly of glomerular origin and mainly composed of plasma albumin.

GLOMERULAR PERMSELECTIVITY

As mentioned earlier, abnormal plasma protein filtration at the glomerular level may be caused by a defect in both size and charge selectivity. Glomerular size-selective dysfunction has been extensively investigated in several kidney diseases by the use of neutral test macromolecules, usually neutral dextrans.^{40,53} These data consistently indicate that permselectivity defects that are responsible for albumin filtration are frequently associated with podocyte foot process effacement and simplification and likely with defective intercellular junctions.⁵⁴

Quantification of the contribution of charge-selectivity defects to proteinuria is more difficult. A few experimental and clinical investigations clearly indicate that proteinuria is indeed associated with abnormal filtration of charged macromolecules,^{55,56} but these data have been questioned because electrically charged test or endogenous macromolecules in the circulation are expected to interfere with other circulating charged solutes and cell membranes.¹ This would result in measured fractional clearance that does not represent effective transport of probe macromolecules. However, even without direct evidence that glomerular membrane charge distribution is altered in proteinuric conditions, the evidence that glomerular albumin filtration is increased without important changes in the fractional clearance of test macromolecules of the same size strongly suggests a role for a charge-selectivity defect in proteinuria of glomerular origin.^{40,56}

Experimental and clinical research has allowed identification of molecular defects underlying some genetic disorders associated with nephrotic syndrome. In Finnish-type nephropathy, a defect in the nephrin gene *NPHS1* is responsible for glomerular dysfunction, proteinuria, and end-stage renal disease.^{3,57} Similarly defects in other genes (*NPHS2*, *LMX1B*, and several others) have been shown to result in defective structure and function of filtration slit proteins or glomerular epithelial cells.⁵⁸ Another condition in which proteinuria is manifest is ischemia-reperfusion injury.⁵⁹ Studies in kidney transplantation and in experimental settings suggest that abnormal elevation of protein in the urine in this condition occurs without major changes in glomerular capillary membrane structure. The evidence indicates⁶⁰ that ischemia per se is responsible for loss of glomerular endothelial glycocalyx, and the previously mentioned effect of fluid shear stress on endothelial cell glycocalyx would reinforce this evidence. Thus abnormal elevation of glomerular protein filtration may derive

from selective changes in ultrastructure and function of membrane components.

TUBULAR HANDLING OF EXCESSIVE FILTERED PROTEINS

EFFECTS OF PROTEIN FILTRATION ON PROXIMAL TUBULE CELLS

An excessive increase in albumin and other plasma protein filtration may result from defective glomerular capillary membrane and/or increase in SNGFR. Both conditions and their combination result in elevated protein concentration in the ultrafiltrate. These filtered proteins are not expected to be entirely reabsorbed by proximal tubule cells because protein reabsorption is believed to operate near maximum under physiologic conditions.

The presence of high-protein concentration within the renal tubule may influence the progression of disease processes. At least two phenomena are expected to occur. The first is related to the fact that if albumin is still present in tubular fluid at the end of the proximal tubule, its concentration increases substantially along the remaining portion of the nephron because of water reabsorption. Thus protein concentration in distal tubules and collecting ducts can reach high values even for a small amount of protein filtered at the glomerular level, with the possibility of these proteins to precipitate and form protein casts.⁶¹ Tubular obstruction may then occur, and entire nephron function is lost, with complete loss of glomerular water filtration.

In addition, structural changes are expected to occur with tubular atrophy, disconnection of the tubule from the Bowman capsule, and glomerular capillary tuft structural changes. This condition is frequently observed in proteinuric kidney diseases at experimental and clinical levels.⁶²

Even before tubular obstruction, important functional changes are expected to occur in proximal tubule cells exposed to abnormal protein concentrations. The protein overload of these cells exposes them to increased workload, and this can lead to loss of reabsorptive capacity due to loss of receptor activity.⁶³ In this condition, the elevated protein concentration along the proximal tubule further increases due to the lower level of absorption and the concomitant water reabsorption. Thus a vicious circle develops, inducing further damage in tubule cells and progressively higher protein concentration along the entire nephron. The consequences of this abnormal glomerular filtration of plasma proteins at both the organ and systemic levels are discussed in the following sections.

RENAL CONSEQUENCES OF PROTEINURIA

GLOMERULAR INJURY

Data from animal models have shown that a wide variety of insults result in a common pathway of glomerular capillary hypertension, increased permeability with excess passage of proteins across the glomerular capillary wall, and progressive glomerular injury⁶⁴ (Fig. 29.5). The key glomerular lesion is sclerosis, characterized by accumulation of extracellular matrix and obliteration of the capillary tuft leading to the loss of renal function.

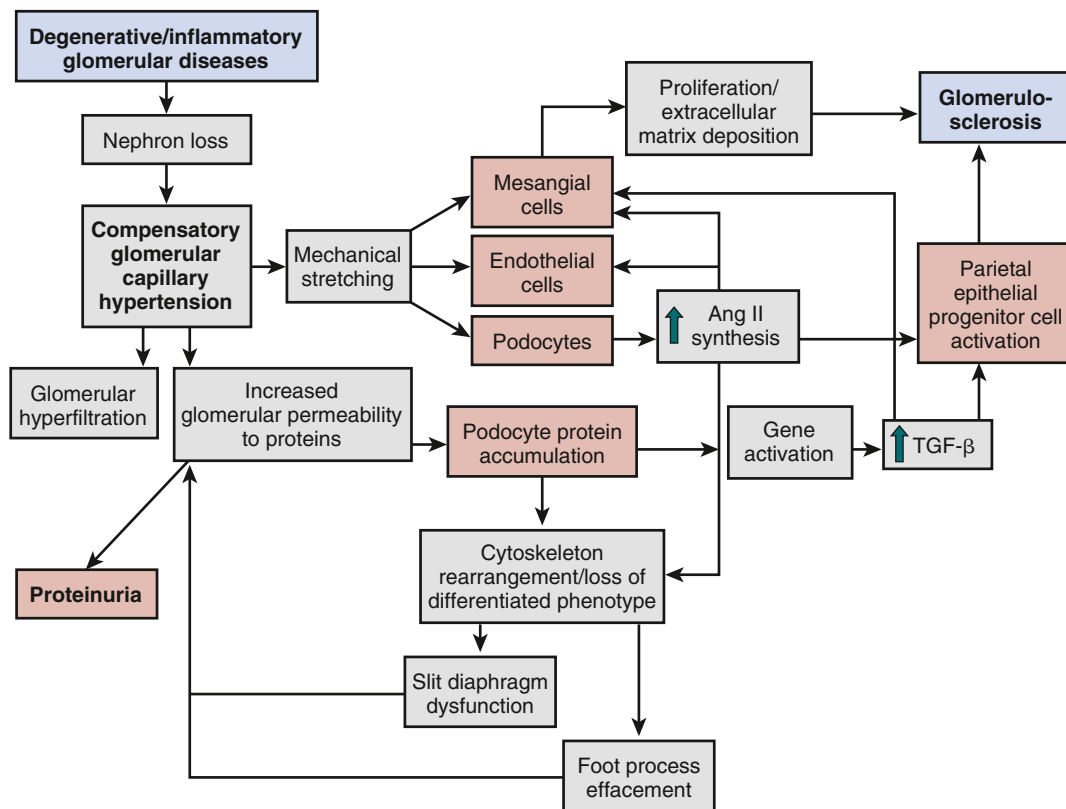


Fig. 29.5 Mechanisms of progressive glomerular injury. A reduction in the number of nephrons as a consequence of various glomerular diseases results in compensatory glomerular hemodynamic changes that are ultimately detrimental. In particular, by mechanical stretching, the increased glomerular capillary pressure directly injures glomerular cells. Glomerular hypertension also impairs the glomerular capillary size-selective function, which causes excessive protein ultrafiltration and eventually podocyte injury and proteinuria.

PODOCYTES: CHANGES IN FUNCTION AND CELL NUMBER

Podocytes show a fairly uniform pattern of response to damage. The intercellular junction and cytoskeletal structure of the foot processes are altered, and the cell shows a simplified, effaced phenotype.^{65,66} These alterations result in the disappearance of the typical slit-diaphragm structures and development of proteinuria. Although podocyte effacement is a hallmark of podocyte disease and nephrotic syndrome, damage to these cells may present as subtle changes that are difficult to quantify.⁶⁷ Major advances in the field of intravital imaging using multiphoton microscopy have enabled us to directly visualize (patho) physiologic processes of the entire glomerulus and the many elements of the glomerular filtration barrier, including podocytes, in unprecedented detail.⁶⁸⁻⁷¹ These new tools are likely to facilitate the next stage in gaining insights to podocyte responses to injury. So far, there is evidence that experimental models of chronic proteinuria, as well as their human counterparts (i.e., minimal-change glomerulopathy, focal and segmental sclerosis, diabetic nephropathy, and membranous nephropathy), have in common ultrastructural findings of severe glomerular epithelial cell damage that include vacuolization, fusion of foot processes, and focal detachment of epithelial cells from the underlying basement membrane.⁷² These changes appear to be the consequence mainly of persistent abnormalities in intraglomerular capillary hemodynamics. Increased capillary hydraulic pressure and flow and activation of local

tissue renin-angiotensin system in podocytes⁷³ eventually impair the size-selective function of the glomerular capillary wall, allowing excess plasma protein to move into the urinary space.⁷⁴ It has been suggested that glomerular permselective dysfunction and albumin filtration depend on the reduced capacity of effaced podocyte foot processes to counteract a transmembrane pressure of 40 mm Hg, with less compression of the GBM.⁷⁵ However, this hypothesis does not account for the role of the subpodocyte space.¹² Evidence from studies that used Munich Wistar Frömter rats, an experimental model of spontaneous proteinuria and glomerulosclerosis,⁷⁶ suggests that the loss of glomerular filtration function depends strongly on the disarrangement of podocytes, which causes expansion of the subpodocyte space, an increase in its hydraulic resistance to filtered water, and increased force acting within the subpodocyte space, which tends to detach podocytes from the capillary membrane.⁷⁷ This hypothesis is further supported by the finding that the podocyte structure is normalized by angiotensin-converting enzyme (ACE) inhibition, which reduces the spatial expansion of the subpodocyte space and ameliorates the hydrodynamic detaching forces that act on podocytes, eventually normalizing water filtration and glomerular barrier function.⁷⁷ Besides being affected by mechanical stress, podocytes are also damaged by excessive protein load resulting from alterations of glomerular permeability to macromolecules. Protein uptake by podocytes may occur through binding to megalin, a receptor for albumin and immunoglobulin (Ig) light chains that is endocytosed after

ligand-binding, as shown in cultured murine podocytes.⁷⁸ Moreover, in vitro exposure of human podocytes to albumin overload prompted an increase in *CUBILIN* and *AMNIONLESS* gene expression, and inhibition of cubulin led to a reduction in albumin uptake, highlighting also the role of the complex cubilin/amnionless receptor-mediated mechanism in albumin endocytosis in these cells.⁷⁹ Mice with protein-overload proteinuria induced by repeated injection of bovine serum albumin developed podocyte injury followed by glomerulosclerosis.⁸⁰⁻⁸² Evidence of a causal link between podocyte protein overload and podocyte damage is provided by studies showing that in rats with renal mass reduction, protein accumulation in podocytes preceded cell dedifferentiation and injury, as characterized by loss of synaptopodin and an increase in desmin expression.⁸³

Podocyte abnormalities were accompanied by upregulation of transforming growth factor- β (TGF- β) mRNA and enhanced production of the related protein.⁸³ In vitro, albumin loading of immortalized mouse podocytes promoted actin-cytoskeleton rearrangement and upregulation of intracellular transduction signals, such as activating protein-1 (AP-1), which is a known stimulus of TGF- β 1 synthesis.⁸⁴

Podocytes possess a complex contractile structure composed of F-actin microfilaments, most abundant in the foot process, connected with adaptor molecules that anchor the slit-diaphragm proteins and $\alpha\beta$ 1 integrins, transmembrane proteins that form focal adhesion complexes and mediate podocyte-GBM matrix interaction.^{85,86} In vitro, actin filament disorganization, as occurs after albumin loading of mouse podocytes,⁸⁴ is closely associated with podocyte shape changes that affect cell adhesion to the extracellular matrix. Podocyte detachment from the GBM likely underlies the decrease in podocyte number in proteinuric glomerular diseases, which has been shown in many experimental and clinical studies.^{87,88} Increased serum levels of indoxyl sulfate, a tryptophan metabolite, have been implicated in the exacerbation of glomerulosclerosis in rat models of proteinuric chronic kidney disease (CKD),^{89,90} by further injuring the podocytes.⁹¹ In the plasma, indoxyl sulfate is largely bound to albumin and localizes to podocytes in uremic rats,⁹² where it exerts its toxic effect. Indeed, mouse and immortalized human podocytes in vitro express aryl-hydrocarbon receptor (AhR), which is activated by binding indoxyl sulfate.⁹¹ This results in reduced expression in myosin/actin cytoskeletal protein and cell adhesion integrin, eventually impairing podocyte viability and decreasing cell number.⁹¹

Apoptosis is considered an additional cause of podocyte loss in proteinuric glomerulopathies. Once detached from the GBM, podocytes become extremely susceptible to apoptosis.⁶⁵ Furthermore, apoptosis may be promoted by locally produced proapoptotic factors. Studies have demonstrated that exogenous TGF- β 1 induced apoptosis in cultured podocytes via the p38 mitogen-activated protein kinase (MAPK) and classic caspase-3 pathways.⁹³ This effect occurred only in wild-type, not p21-null cultured podocytes, indicating that the cyclin-dependent kinase (CDK) inhibitor p21 is required for TGF- β 1-induced apoptosis.⁹⁴ Of note, like TGF- β 1, p21 is increased in podocytes in experimental models of membranous nephropathy⁹⁵ and diabetic nephropathy.⁹⁶ In summary, protein accumulation in

podocytes induces TGF- β 1 production, leading to podocyte apoptosis. Podocytes maintain their physiologic homeostasis through a high level of basal autophagy, a process that helps to reduce cell damage and promote cell survival.⁹⁷ However, at least in patients with diabetic kidney disease, podocyte autophagy is impaired, exacerbating injury and proteinuria.^{98,99} Of note, in injured podocytes, METTL14, a protein involved in RNA modification, promotes the degradation of sirtuin 1 (*Sirt1*) mRNA m6A, a signaling pathway known to enhance podocyte autophagy.¹⁰⁰

Evidence indicates that angiotensin II (Ang II) contributes to perpetuate podocyte injury in proteinuric nephropathies, eventually promoting progression to end-stage kidney disease.¹⁰¹ Mechanical strain increases Ang II production and expression of angiotensin II type 1 (AT₁) receptors in podocytes,⁷³ potentially contributing to further sustaining the glomerular hypertension-induced damage in CKD. However, there is also evidence that independent of its hemodynamic effect, Ang II may directly impair the glomerular barrier-sieving function, possibly through inhibition of nephrin expression by podocyte, the essential protein component of the glomerular slit diaphragm.^{102,103} This observation has been confirmed in studies in diabetic animals showing that blockade of Ang II synthesis/activity preserved the expression of nephrin in the glomeruli and prevented overt proteinuria.^{104,105} Thus at least in diabetes, a pathogenetic relationship between Ang II and early proteinuria via functional podocyte alteration through modulation of nephrin protein level has been suggested. Recently, it has also been shown that in murine podocytes Ang II impaired glycolysis by reducing the expression of pyruvate kinase M2, a key glycolytic enzyme, accompanied by cytoskeletal remodeling and podocyte apoptosis.¹⁰⁶ Thus insufficient energy supply to the foot process sustained by Ang II leads to podocyte injury. Additionally, pyruvate kinase M2 expression was found to be reduced in podocytes from kidney biopsies of patients with diabetic kidney disease,¹⁰⁶ further suggesting that impaired glycolysis is an additional process contributing to podocyte damage. Moreover, in the setting of diabetes, after the initial insult of hyperglycemia and intraglomerular hypertension, Ang II plays a relevant role to sustain glomerular injury via persistent activation of Notch1 and Snail signaling in the podocyte, eventually resulting in persistent downregulation of nephrin expression.¹⁰⁷ The consistency of these findings in diabetic Zucker diabetic fatty (ZDF) rats with overt nephropathy and in type 2 diabetic patients with established nephropathy provides robust insight to implicate an important role for the Ang II Notch1/Snail axis in perpetuating podocyte damage.

MicroRNAs (miRNAs) are a class of short (21–24 nucleotides) noncoding RNAs that regulate gene expression through posttranslational and epigenetic mechanisms, thereby affecting several cellular processes from development to disease conditions.¹⁰⁸ miRNAs are critical players in podocyte homeostasis (Fig. 29.6) because targeted deletion of the helicase with RNase motif Dicer or the class II ribonuclease III enzyme Drosha in these cells leads to proteinuria and glomerulosclerosis.¹⁰⁹⁻¹¹² Moreover, several miRNAs have been found to be dysregulated in podocyte injury, a pathologic mechanism causing glomerular injury and sclerosis. Because mature podocytes must withstand fluctuating pressures and potentially harmful molecules contained in the primary filtrate, they are unlikely to be

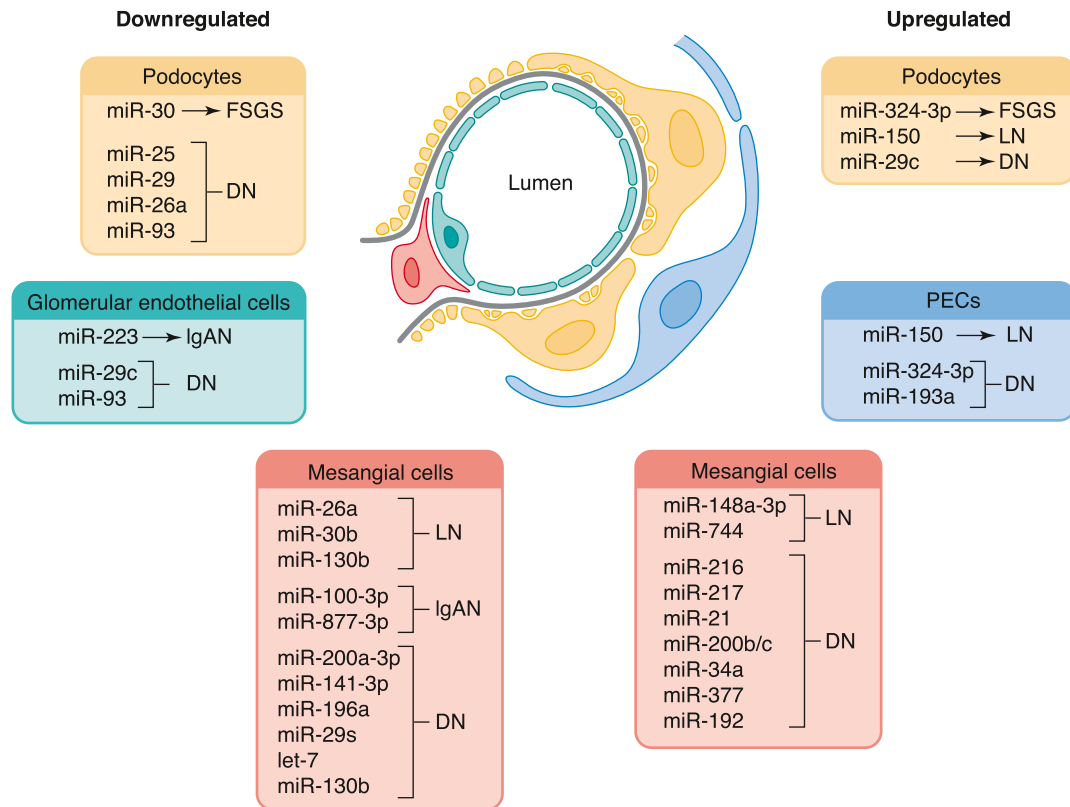


Fig. 29.6 miRNA dysregulation in glomerular disease. Changes in miRNAs in different populations of glomerular cells (podocytes, PECs, glomerular endothelial cells, and mesangial cells) occurring in FSGS, lupus nephritis, IgA nephropathy, and diabetic nephropathy. *DN*, Diabetic nephropathy; *FSGS*, focal and segmental glomerulosclerosis; *IgAN*, IgA nephropathy; *PECs*, parietal epithelial cells. (From Trionfini P, Benigni A. MicroRNAs as master regulators of glomerular function in health and disease. *J Am Soc Nephrol*. 2017;28:1686–1696.)

static structures.¹¹³ Remodeling of the actin cytoskeleton plays a primary role in the structural adaptations made by podocytes to preserve their glomerular filtration properties. The podocyte cytoskeleton is finely regulated by a set of miRNAs that are expressed mainly in the adult kidney, such as miR-30, miR-132, miR-134, and miR-29a. miR-30 family members are highly expressed in human podocytes and, in addition to protecting them against apoptosis,¹¹⁴ they promote podocyte actin fiber stability by controlling calcium/calcineurin signaling through the inhibition of several components of this pathway.^{115,116} Dysregulation of calcium/calcineurin signaling leads to podocyte cytoskeletal damage, which is a key feature of various glomerular diseases, such as focal and segmental glomerulosclerosis (FSGS), characterized by the early onset of podocyte injury.^{115,117} It is notable that miR-30s are significantly downregulated in the podocytes of patients with FSGS and in the rat model of FSGS induced by the podocyte toxin puromycin aminonucleoside. Furthermore, fibrogenic factors, such as TGF- β , reduce miR-30 expression in vivo and in cultured podocytes.^{114,118} An additional study underlines the role of miRNAs in regulating podocyte cytoskeletal dynamics. It showed that brain-derived neurotrophic factors can repair podocyte damage in vitro and in a mouse model of FSGS by inducing miR-132 and inhibiting miR-134 expression upon binding to its receptor on podocytes. Brain-derived neurotrophic factor-induced modulation of miR-132 and miR-134 has been found to be essential for increasing actin polymerization, favoring foot process elongation

that contrasts the cell flattening induced by proteinuric conditions.¹¹⁹

A crosstalk with proximal tubule cells has been suggested to contribute to podocyte function through the release of nicotinamide mononucleotide (NMN), which might have implications for persistent podocyte dysfunction in proteinuric diseases.¹²⁰ In a mouse model, diabetes-induced down regulation of Sirtuin1 (Sirt1), a highly conserved protein deacetylase in proximal tubules.¹²¹ The low Sirt1 expression reduces the release of nicotinamide mononucleotide by tubule cells, eventually decreasing local NMN concentrations. In vitro studies have shown that in the absence of NMN, the expression of the tight junction protein Claudin-1 in podocytes is no longer silenced.¹²⁰ Claudin-1 is reported to activate the intracellular β -catenin-Snail pathway,¹²² which eventually leads to glomerular barrier dysfunction through downregulating synaptopodin or podocin expression in podocytes.¹²³

MESANGIAL CELLS: PROLIFERATION AND DEPOSITION OF EXTRACELLULAR MATRIX

Because it is close to the capillary lumen, the mesangium may be exposed to macromolecules crossing the endothelial layer, although under normal conditions, they do not accumulate.¹²⁴ However, in rats having undergone unilateral nephrectomy¹²⁵ or in puromycin-aminoglycoside-induced nephrosis,¹²⁶ intravenous infusion of colloidal carbon leads to the accumulation of the macromolecular tracer in the mesangial space. To prevent accumulation of proteins,

mechanisms exist for their effective removal. These include transport along the mesangial stalk in cleftlike spaces, as well as phagocytosis and degradation by mesangial cells.¹²⁷

It has been shown that IgG and IgA can be taken up by both receptor-independent and receptor-mediated processes.¹²⁸ Another important factor for clearance of Igs from the mesangium may be complement factor D, a serine protease essential for activation of the complement system through the alternative pathway, which is constitutively expressed within the glomerulus.¹²⁹ Interestingly, mice deficient in complement factor D spontaneously develop a mesangial immune-complex deposition disease associated with albuminuria.¹³⁰ This indicates that complement factor D is necessary to prevent mesangial accumulation of immunoglobulin deposits.

Whether abnormal local accumulation of proteins promotes mesangial cell proliferation and mesangial matrix deposition remains, however, ill-defined. The mesangial cell is a critical part of the glomerular functional unit interacting closely with endothelial cells and podocytes.^{129,131} Alterations in one cell type can produce changes in the others. As such, key survival factors for mesangial cells including platelet-derived growth factor-B (PDGF-B) are generated by endothelial cells, and mesangiolysis has been shown in knockout mice lacking endothelial PDGF-B.¹²⁹ Whether cytokines/chemokines generated by podocytes also influence mesangial cells has yet to be clearly defined, but the observation that podocyte injury frequently results in mesangial cell proliferation supports the existence of such cytokine crosstalk.¹²⁹

Besides PDGF-B, other growth factors shown to influence mesangial cell proliferation and mesangial matrix accumulation include PDGF-C, fibroblast growth factor, hepatocyte growth factor, epidermal growth factor (EGF), connective tissue growth factor, and TGF- β .¹²⁹ Effects of vasoactive hormones such as angiotensin II on mesangial cell proliferation may be mediated indirectly through the generation of growth factors such as EGF.¹³² In rats with renal mass ablation, glomerular TGF- β 1 upregulation is associated with phenotypic transformation of mesangial cells.⁸³ At least in experimental diabetes, increased matrix production in mesangial cells is induced by a fine balance between the upregulation of miR-192, miR-200b/c, miR-216a, and miR-377¹³³ and the downregulation of miR-29s¹³⁴ and let-7.¹³⁵ Notably, in mesangial cells, TGF- β induces signaling loops that amplify and create a chronic state of profibrotic pathway activation, modulating the expression of miR-192, miR-200s, miR-21, and miR-130b.¹³⁶⁻¹³⁸ In addition, sodium-glucose cotransporter 2 (SGLT2) is expressed in mesangial cells of dB/dB mice and increased in vitro approximately fivefold upon exposure to high glucose concentration.¹³⁹ The increased glucose uptake, as well as Na⁺ through SGLT2 by mesangial cells, activates intracellular pathways, including diacylglycerol protein kinase C pathway,¹⁴⁰ resulting in swelling, dysfunction, and loss of cells.¹⁴¹ Of note, a low dose of the SGLT2 inhibitor canagliflozin normalized TGF- β 1 and fibronectin mRNA levels in the mesangial cells of dB/dB mice.¹³⁹

ENDOTHELIAL CELLS: APOPTOSIS

Glomerular endothelial injury is a common feature of many human diseases, such as diabetic nephropathy, hypertension, thrombotic microangiopathy, and preeclampsia. Evidence

has been provided that there is close crosstalk of podocytes with glomerular endothelial cells, a key interaction for the normal function of the glomerular capillary barrier.^{142,143} The final steps in glomerular endothelial cell differentiation involve the formation of fenestrae, plasma membrane-lined circular pores that perforate the flattened glomerular endothelium.¹⁴⁴ The fenestrated phenotype of glomerular endothelial cells is induced by vascular endothelial growth factor (VEGF), a molecule constitutively expressed and secreted by podocytes.^{142,144} Podocyte-produced VEGF regulates the structure and function of adjacent glomerular endothelial cells by binding to VEGF receptors 1 and 2 (VEGFR1 and VEGFR2) and neuropilin-1/2.¹⁴⁵ Moreover, deletion of the podocyte-specific transcription factor LMX1B in mice, which results in the loss of many features of podocyte differentiation, including VEGF-A expression, is associated with failure of glomerular endothelium to differentiate and develop fenestrae.¹⁴⁶ Thus loss of podocytes secondary to protein-induced cell injury may lead to reduced VEGF production influencing glomerular endothelial fenestrae formation and eventually leading to endothelial cell apoptosis.¹⁴⁷ Conversely, in vitro evidence has shown that blockade of VEGF in glomerular endothelial cells enhanced the release of endothelin-1 (ET-1), which induced nephrin shedding from podocytes,¹⁴⁸ leading to further glomerular protein permeability dysfunction.

Moreover, ET-1 activates podocytes to release heparanase.¹⁴⁹ In mice, podocyte-specific deletion of the endothelin receptor prevented the diabetes-induced increase in glomerular heparanase expression and consequent reduction in heparan sulfate expression, endothelial glycocalyx thickness, and development of proteinuria that was observed in wild-type mice.¹⁴⁹ Heparanase-deficient mice were resistant to the development of proteinuria and renal damage on induction of type 1 diabetes mellitus.¹⁵⁰

Taken together, these studies indicate that the toxic effects of excess ultrafiltered plasma proteins on podocytes may alter podocyte-endothelial interaction, thereby further enhancing glomerular permeability to proteins through a complex interplay of molecular signaling.

PARIETAL EPITHELIAL CELLS: ACTIVATION

Changes in glomerular permselective function, as occur in proteinuric glomerulopathies, elevate the filtered load of plasma albumin and consequently its concentration in the Bowman space (Fig. 29.7). Evidence has been provided that the abnormally filtered albumin impairs the mechanism underlying regeneration of protein-induced damage to podocytes.¹⁵¹ Glomerular injury caused by multiple etiologies can lead to activation and accumulation of parietal epithelial cells (PEC) within the Bowman space as a common response to damage,¹⁵² as shown in several human proliferative glomerulonephritides. Although extracapillary proliferation is a relatively straightforward pathologic change to recognize, determining its cellular components has been more controversial. Traditional concepts, largely from immunohistochemical studies, indicate that the multilayered cellular lesions are a mixture of glomerular PECs, macrophages, and myofibroblasts.¹⁵³⁻¹⁵⁵ In both animal models and human tissues, PECs predominate when the Bowman capsule is intact. A heterogeneous population of renal progenitor cells, previously identified

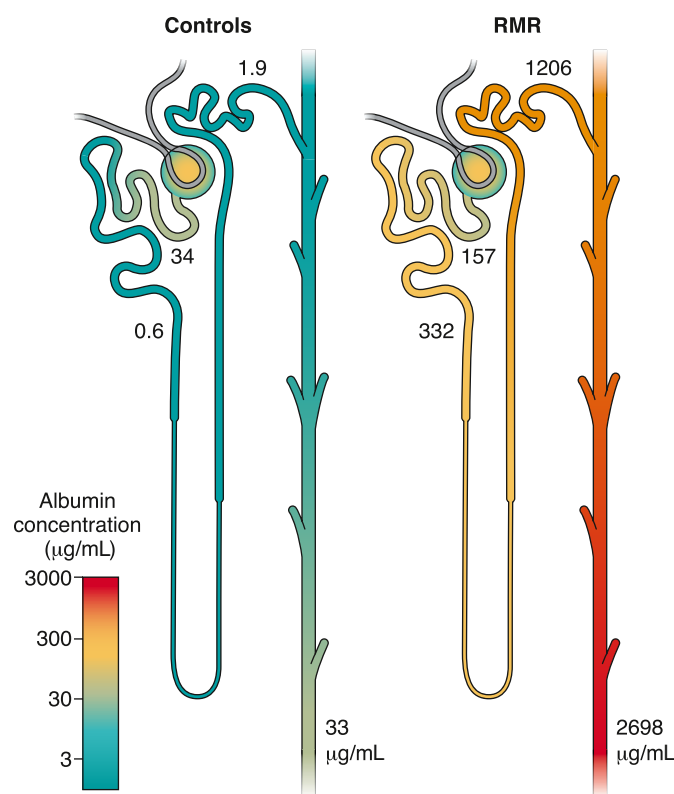


Fig. 29.7 Estimated albumin concentrations along the nephron. Color-coded graphical representation of estimated albumin concentration along the entire nephron in the two animal groups (control group and renal mass reduction group). Numbers represent local group average albumin concentration in $\mu\text{g/mL}$. **RMR**, Renal mass reduction. (Modified from Sangalli F, Carrara F, Gaspari F, et al. Effect of ACE inhibition on glomerular permselectivity and tubular albumin concentration in the renal ablation model. *Am J Physiol Renal Physiol*. 2011;300:F1291–F1300.)

in normal human Bowman capsule,¹⁵⁶ has been documented in hyperplastic lesions of human crescentic glomerulonephritis.¹⁵⁷ The extracapillary lesions could therefore be the result of dysregulated proliferation of renal progenitor cells in response to injured podocytes.¹⁵⁷ This possibility is supported by findings in Munich Wistar Fromter rats, which are genetically programmed to develop renal damage characterized by excessive progenitor cell migration and proliferation leading to their accumulation into cellular lesions and glomerulosclerosis.¹⁵⁸ Among the factors that influence the maladaptive PEC response, ultrafiltered albumin has been regarded as a critical player that impairs podocyte regeneration.¹⁵⁹ Circulating components of the complement system are also lost in the urine in proteinuric conditions and become activated at the glomerular level, favoring progression of the lesions.^{83,160–162} Thus abnormal fixation of ultrafiltered complement C3 is detected in podocytes showing signs of dedifferentiation and injury during the early stage of proteinuric disease in rats with remnant kidney,⁸³ in mice with protein-overload proteinuria,¹⁶² and in BTBR ob/ob mice.¹⁶³ We also demonstrated that complement activation via alternative pathway is a pivotal trigger for podocyte loss and PEC activation and migration to capillary tuft, leading to glomerulosclerosis in the model of protein-overload

proteinuria, as indicated in factor H (*Cfh*^{-/-})– or factor B–deficient mice compared with wild-type littermates.¹⁶⁴ In protein-overload mice, PEC dysregulation was associated with upregulation of CXCR4 and GDNF/c-Ret axis. In patients with proteinuric nephropathy, glomerular C3/C3a paralleled PEC activation, CXCR4 and GDNF upregulation. These results indicate that mechanistically uncontrolled alternative pathway complement activation is not dispensable for podocyte-dependent PEC activation resulting in glomerulosclerosis.¹⁶⁴

PECs expressing the progenitor cell marker CD133⁺CD24⁺ have also been reported to proliferate and accumulate into multilayered cellular lesions in patients with glomerulonephritides characterized by extracapillary proliferation but not in nonproliferative nephropathies, such as membranous or diabetic nephropathies.¹⁶⁵ Upregulation of the CXCR4 chemokine receptor on these progenitor cells was accompanied by high expression of its ligand, SDF-1, in podocytes.¹⁶⁵ Moreover, parietal epithelial cell proliferation was associated with increased expression of the Ang II subtype 1 receptor. Renin-angiotensin system blockade normalized CXCR4 and Ang II subtype 1 receptor expression on parietal progenitor cells concomitant with regression of crescentic lesions. Together, these findings suggest that the glomerular hyperplastic lesions derive from the proliferation and migration of renal progenitors in response to injured podocytes and that the Ang II/AT₁ receptor pathway may contribute, along with the SDF-1/CXCR4 axis, to the dysregulated response of parietal epithelial cell precursors. Macrophage migration inhibitory factor (MIF), a pleiotropic proinflammatory cytokine produced by injured podocytes,¹⁶⁶ also mediates pathologic PEC proliferation via the CD74/CD44 receptor complex. There is evidence that MIF, as well as its receptor complex CD74/CD44, is upregulated in the glomeruli of patients and mice with proliferative glomerulonephritides.¹⁶⁷ Moreover, increased albumin levels in initial urine induce CD44 expression in PEC via activating the megalin-mediated extracellular signal-regulated kinase (ERK) signaling pathway.¹⁶⁸ Thus the MIF-CD74/CD44 signaling pathway suggests a further potential mechanism for PEC pathophysiologic proliferation and crescent lesion formation.

Parietal epithelial cell activation is increasingly recognized and seems to be present also in most forms of focal segmental glomerulosclerosis (FSGS), characterized by nephrotic syndrome and often leading to a progressive decline in renal function.¹⁶⁹ Using a lineage tagging approach in mice, it has been documented that activated PECs invade the affected segment of the capillary PECs via cellular adhesions between the Bowman capsule and capillary tuft, and produce extracellular matrix, contributing to glomerulosclerosis.¹⁷⁰ A Notch signaling pathway has been proposed to play a role in orchestrating parietal epithelial cell phenotypic changes in FSGS.¹⁷¹ This possibility rests on in vitro evidence that in cultured mouse PECs, TGF- β enhanced Notch mRNA expression, which resulted in a significant upregulation of target genes associated with mesenchymal cell phenotype, such as α -smooth muscle actin, vimentin, and Snail.¹⁷¹

miRNAs have emerged as important regulators of gene expression in PECs.¹⁰⁸ Indeed, robust miR-150 expression in PECs and podocytes of patients with lupus nephritis correlated positively with disease chronicity scores and

the expression of profibrotic proteins. Increased miR-150 would foster the production of profibrotic molecules through the downregulation of its predicted target, SOCS1. The latter protein acts as a negative regulator of the JAK/STAT signaling pathway, which promotes the transcription of genes involved in cell proliferation, inflammation, and fibrosis.^{172,173} Similarly to miR-150, we identified miR-324-3p increased expression in PECs, as well as podocytes in an FSGS rat model. Increased expression of miR-324-3p was associated with the downregulation of its target propyl endopeptidase—involved in the formation of the antifibrotic peptide Ac-SDKD—in fibrotic areas of the kidneys of diseased rats.¹⁷⁴ Human PECs isolated from naïve Bowman capsules express significant levels of miR-193a, which works as a suppressor of podocyte differentiation by inhibiting the expression of the transcription factor Wilms tumor protein (WT1).^{175,176} The downregulation of miR-193a is associated with PEC transdifferentiation toward a podocyte phenotype, whereas its overexpression leads to their abnormal activation, a prerequisite for the formation of crescents in proliferative glomerulonephritis.¹⁷⁶ In line with this, isolated glomeruli from individuals with FSGS are characterized by increased expression of miR-193a, compared with normal kidneys or kidneys affected by other glomerular diseases.¹⁷⁵ All of the previously described evidence, together with findings that the number of crescents was reduced by anti-miR-193a in a mouse model of nephrotoxic nephritis,¹⁷⁶ concur to indicate that miR-193a is a promising therapeutic target for FSGS. Moreover, the kidneys of patients with ANCA-associated crescentic glomerulonephritis and mice with nephrotoxic nephritis are characterized by the upregulation of miR-155, which drives the Th17 immune response and tissue injury.¹⁷⁷

LOSS OF GLOMERULAR CAPILLARIES: POSTGLOMERULAR HYPOXIA

Irrespective of the underlying process leading to glomerular endothelial damage, such as increased intracapillary pressure and/or podocyte loss, the net result is rarefaction of glomerular capillaries. Loss of glomerular capillary loops translates into diminished postglomerular blood flow from affected glomeruli and downstream injury of the peritubular capillary network. Microvascular dysfunction causes progressive scarring of renal tissue by creating a hypoxic environment that triggers a fibrotic response in tubulointerstitial cells.¹⁷⁸ This, in turn, has an impact on adjacent unaffected capillaries and glomeruli, further extending the hypoxic area and leading to a vicious cycle of progressive destruction of the kidney and decline of renal function to end-stage organ failure.

Indeed, in animal models of proteinuric CKD including anti-Thy1 glomerulonephritis, 5/6 remnant kidney, diabetic nephropathy, and doxorubicin (Adriamycin)-induced nephrosis, the immunohistochemical detection of hypoxia-dependent pimonidazole protein adducts has revealed that renal tissue hypoxia is present early in the course of the disease.¹⁷⁹ Although data from animal models provide a compelling argument for postglomerular hypoxia in proteinuric diseases as a primary mediator of progressive renal scarring, data in humans are scarce. Nevertheless, that hypoxia-related injury also applies to humans may be deduced by the finding that there is increased expression

of hypoxia-inducible factor (HIF)—a key regulator of the adaptive response to hypoxia controlling expression of hundreds of genes¹⁸⁰—in biopsies of patients with diabetic nephropathy, IgA nephropathy, and chronic allograft nephropathy.¹⁸¹

TUBULAR INJURY

Glomerular ultrafiltration of excessive amounts of plasma protein-associated factors incites tubulointerstitial damage and further promotes the effects of glomerular disease on the tubular compartment. Noxious substances in the proteinuric ultrafiltrate may set off tubular epithelial injury with tubular apoptosis, secondary generation of inflammatory mediators, and peritubular inflammation.⁴ The mechanisms whereby increased urinary protein concentration leads to nephrotoxic injury are multifactorial and involve complex interaction between numerous pathways of cellular damage (Fig. 29.8).

TUBULE CELLS: APOPTOSIS AND TUBULO GLOMERULAR DISCONNECTION

Evidence suggests that proteinuria causes tubule cell apoptosis. In cultured proximal tubule cells, delipidated albumin induced apoptosis in a dose- and time-dependent manner,¹⁸² as characterized by internucleosomal DNA fragmentation, morphologic changes including cell shrinkage and nuclear condensation, and plasma membrane alterations.¹⁸² Kidneys of rats with albumin overload proteinuria¹⁸³ or passive Heymann nephritis⁶² showed increased numbers of terminal dUTP nick-end labeling-positive apoptotic cells in the tubulointerstitial compartment. In tubules, most of the positive cells expressed angiotensin II subtype 2 (AT2) receptors.¹⁸³ Findings of reduced phosphorylation of ERK and Bcl-2 suggested an AT2 receptor-mediated mechanism underlying tubular cell apoptosis.¹⁸³

Apoptotic cells expressing both proximal and distal tubular phenotype were detected in biopsy specimens from patients with primary focal segmental glomerulosclerosis, with strong positive correlation between proteinuria and incidence of tubular cell apoptosis.¹⁸⁴

Renal proximal tubule cells have a remarkable ability to reabsorb large quantities of albumin through clathrin- and megalin receptor-mediated endocytosis.⁶³ Megalin is the sensor that determines whether cells will be protected from or injured by albumin. It has been shown that megalin binds the serine/threonine kinase PKB, crucial for the phosphorylation of Bad, the Bcl2-associated death promoter.¹⁸⁵ Low concentrations of albumin lead to activated PKB and phosphorylation of the Bad protein, which inhibits apoptosis.¹⁸⁶ On the other hand, overload of albumin leads to decrease in megalin expression on the plasma membrane of proximal tubule cells that is associated with reduction of PKB activity and Bad phosphorylation.¹⁸⁷ The result is albumin-induced apoptosis. Recent evidence in experimental mice has also shown that elevated plasma levels of proprotein convertase subtilisin/kexin type 9 (PCSK9), a regulator of cholesterol, filtered by glomeruli, bind to and downregulate megalin in the proximal tubule. This reduces its function, thereby attenuating megalin-driven protein reabsorption.¹⁸⁸ While PCSK9-induced decrease in megalin

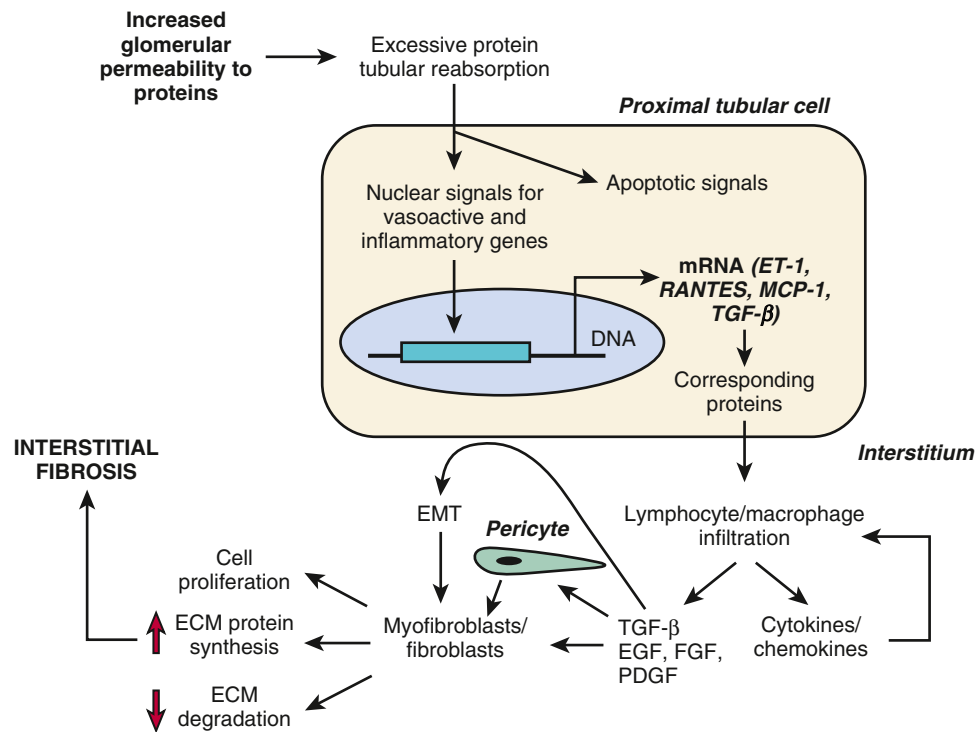


Fig. 29.8 Mechanisms of tubulointerstitial damage induced by proteins. Protein overload of proximal tubule cells as a consequence of increased glomerular permeability to proteins activates intracellular signals that promote cell apoptosis or cause increased production of inflammatory and vasoactive mediators, and growth factors. These substances are released into the interstitium, inducing progressive inflammation and injury. MCP-1, monocyte chemoattractant protein-1, RANTES, regulated upon activation, normal T cell expressed and secreted. ET-1, endothelin-1. TGF- β , transforming growth factor- β . EGF, epidermal growth factor. FGF, fibroblast growth factor. PDGF, platelet-derived growth factor. EMT, epithelial-to-mesenchymal transdifferentiation. ECM, extracellular matrix.

aggravates proteinuria, it remains to be determined whether it also translates into tubular cell apoptosis.

In cultured proximal tubule cells, albumin repletion with fatty acids and its association with linoleic acid induced more apoptosis than the exposure to defatted albumin alone.¹⁸⁹ Furthermore, another study showed that nondelipidated albumin or albumin conjugated with palmitate, but not fatty acid-free albumin, altered both tubule mitochondrial viability and membrane potential and caused cytochrome *c* release.¹⁹⁰ In concert with the decline of mitochondrial parameters, fatty acid overload led to a redox imbalance, which deactivated the antioxidant protein peroxiredoxin 2 and caused peroxide-mediated apoptosis through the redox-sensitive pJNK/caspase-3 pathway. These data were taken to suggest that attempts at lowering circulating fatty acid levels may be important in both preserving redox balance and limiting tubule cell damage.¹⁹⁰ An additional biochemical mechanism has been proposed linking lipotoxicity to tubule apoptosis in proteinuric conditions.¹⁹¹ Starting from the concept that diseased glomeruli with impaired permselectivity allow filtration and proximal tubule reabsorption of nonesterified fatty acids (NEFAs) bound to albumin, it was shown that an accumulation of metabolites of NEFA, long-chain acyl-CoA could stimulate lipoapoptosis by competing with the structurally similar phosphoinositide phosphatidylinositol 4,5-bisphosphate [PI(4,5)P2] for binding to Na⁺/H⁺ exchanger NHE1, a regulator for proximal tubule cell survival, thus interrupting PI(4,5)P2 prosurvival activity.¹⁹⁰

Proximal tubule cell apoptosis was found to contribute to tubulo-glomerular disconnection and atrophy in response to proteinuria in animal models of proteinuric nephropathies.⁶² Injured or dying cells release molecules, referred to as danger-associated molecular patterns (DAMPs).^{192,193} They trigger inflammation by engaging pattern recognition receptors such as Toll-like receptors (TLR)¹⁹⁴ and nucleotide-binding domains, leucine-rich repeat-containing proteins (NLRs).¹⁹⁵ Thus besides promoting tubulo-glomerular disconnection, proximal tubular cell apoptosis contributes to create a local proinflammatory microenvironment.

TUBULE CELLS: ACTIVATION

Disruption of glomerular filtration barrier results in the spillage into early urine of a mixture of many plasma proteins such as albumin, immunoglobulin, transferrin, α 2-macroglobulin, and complement C3. Their relative role in proximal tubular cell activation and injury is still debated.¹⁹⁶ This particularly refers to albumin, whose different modified forms, such as glycated, carbamylated, and others,¹⁹⁷ are processed or edited differently from nonmodified albumin after proximal tubule cell uptake.¹⁹⁸ Albumin reabsorbed by apical megalin-cubulin receptor-mediated endocytosis of clathrin-coated vesicles or clathrin-independent endocytosis is released from the megalin-cubulin complex in the acidic environment of endosomes.¹⁹⁸ Next, nonmodified albumin binds to the neonatal Fc receptor (FcRn) and can be transcytosed at the basolateral side of the cell. By contrast, glycated,

carbamylation, and oxidized albumin derivatives do not bind to FcRn and accumulate into lysosomes, where albumin undergoes partial degradation.¹⁹⁸ Albumin fragments are then released in the urine by exocytosis, a process aimed to eliminate potentially toxic forms of albumin. However, the fact that this proximal tubular albumin sorting pathway is progressively lost with the progression of proteinuric kidney disease¹⁹⁹ and the evidence of direct effects of modified albumin on inflammatory molecules^{200–202} suggest that any form of albumin could eventually lead to proximal tubular cell activation and injury. Nonetheless, there is evidence that receptor-mediated endocytosis of excessive proteins, including albumin, at the apical pole of the proximal tubule cells is associated with phenotypic changes characteristic of an activated state.

Insights into specific mechanisms linking protein uptake to cell activation have come from *in vitro* studies using polarized proximal tubule cells to assess the effect of apical exposure to proteins. Collectively, they show that protein overload induces a proinflammatory phenotype.^{203–206} Indeed upregulation of inflammatory and fibrogenic genes and production of related proteins have been reported following a challenge of proximal tubule cells with plasma proteins. They include cytokines and chemokines, such as monocyte chemoattractant protein-1 (MCP-1); regulated upon activation, normal T cell expressed and secreted (RANTES); interleukin-8 (IL-8); and fractalkine.^{203–206} Moreover, levels of the profibrogenic cytokine TGF- β and its type I receptor,²⁰⁷ tissue inhibitors of metalloproteinase (TIMP)-1 and TIMP-2, as well as membrane surface expression of the $\alpha\beta 5$ integrin,²⁰⁸ were also highly increased *in vitro* upon stimulation by plasma proteins.

Investigations of the molecular mechanisms underlying chemokine and growth factor upregulation in proximal tubule cells on protein challenge have focused on the activation of transcription factor NF- κ B.²⁰³ Other studies confirmed this pathway^{209,210} and revealed reactive oxygen species as second messengers.^{205,211}

Extrapolation from such *in vitro* data to the human situation may be difficult considering the conflicting data observed with different proteins in different cell systems,²¹² as well as the reported changes in the expression of several genes of unknown function.²¹³ However, the *in vitro* observations have recently been confirmed through transcription analysis by cDNA microarray of renal proximal tubule epithelial cells isolated by laser capture microdissection from patients with proteinuric nephropathies.²¹⁴ More than 160 genes including those encoding for signal transduction, transcription and translation, and apoptotic and inflammatory proteins were identified as being regulated differently from those in proximal tubule cells from control subjects.

Evidence implicates megalin as a central element of the signaling pathway linking protein reabsorption and gene regulation in proximal tubule cells.²¹⁵ Megalin is subjected to regulated intramembrane proteolysis (RIP), an evolutionarily conserved process linking receptor function with transcriptional regulation.²¹⁶ The C-terminal fragment is released into the cytosol and translocates to the nucleus, where it interacts with other proteins to regulate expression of specific genes.²¹⁵ This function may explain the phenotypic change of proximal tubules in proteinuric kidney disease.

That megalin contributes to the early activation of proximal tubule cells in nonselective proteinuria has been documented in megalin-knockout/*NEP25* mice given immunotoxin LMB2, a model for nephrotic syndrome, focal segmental glomerulosclerosis, and tubulointerstitial injury.²¹⁷ Megalin-deficient proximal tubule cells reabsorbed less protein *in vivo* and expressed less tubular injury markers, such as MCP-1 and heme-oxygenase 1.²¹⁷

The proteinuric ultrafiltrate may also activate TLRs and promote an inflammatory response.²¹⁸ It has been demonstrated that proximal tubule epithelial cells were sites of robust expression of TLR-9 mRNA and protein—a receptor for CpG DNA—in NZBxNZW lupus mice with overt nephropathy.²¹⁹ Upregulation of TLR-9 expression was accompanied by the development of proteinuria and correlated with tubulointerstitial damage. Furthermore, abundant TLR-9 staining in proximal tubule cells of lupus patients correlated with tubulointerstitial damage. Thus tubular TLR activation may occur due to filtration of plasma proteins, which include immune complexes containing DNA enriched in CG motifs.^{220,221}

The proximal tubule bears other receptors for ultrafiltered proteins, such as cytokines and growth factors.⁴ Usually these molecules are present in high-molecular-weight precursor forms or bound to specific binding proteins, which regulate their biological activity. They can be found in nephrotic tubular fluid. Proximal tubular fluid of proteinuric rats may contain insulin-like growth factor I (IGF-I)²²² or hepatocyte growth factor (HGF).²²³ Under physiologic conditions the high molecular weight of TGF- β complexes prevents glomerular ultrafiltration of this pluripotent cytokine. However, in proteinuric glomerular diseases, TGF- β is present in early proximal tubular fluid and at least a portion is bioactive.²²³ IGF-I, HGF, and TGF- β are also present in the urine of patients with proteinuric diseases.²²⁴

Collectively, the tubular response to these growth factors can be described as activation or as a moderate change toward a cell phenotype resembling cell injury, which includes a moderate increase in collagen and fibronectin.^{222,225} With proteinuria, a putative key factor in tubule cell activation and damage is the excess glomerular filtration of serum-derived complement C3, the central molecule in the complement system that exerts proinflammatory potential.²²⁶ Renal tubule epithelial cells appear most susceptible to luminal attack by the C5b-9 membrane attack complex because of the relative lack of membrane-bound complement regulatory proteins such as membrane cofactor protein (CD46), decay-accelerating factor (CD55), and CD59 on the apical surface.²²⁷ In rats with severely reduced renal mass^{228,229} or protein-overload proteinuria,¹⁶² C3 colocalized with proximal tubule cells engaged in high protein uptake. By limiting the transglomerular passage of proteins, treatment with ACE inhibitors was an effective maneuver to reduce C3 load of tubule cells in remnant kidneys.²²⁸ C3 and other complement proteins are also found in proximal tubules in renal biopsy material from nephrotic patients.^{226,230}

The injurious role of plasma-derived C3, as opposed to tubular cell-derived C3, has been documented in C3-deficient kidneys transplanted into wild-type mice.¹⁶² Protein overload led to the development of glomerular injury, accumulation of C3 in proximal tubules, and tubulointerstitial changes. Conversely, when wild-type

kidneys were transplanted into C3-deficient mice, protein overload led to a milder disease and abnormal C3 deposition was not observed. Thus ultrafiltered C3 contributes more to tubulointerstitial damage induced by protein overload than locally synthesized C3.

INTERSTITIAL INFLAMMATION AND INJURY

In proteinuric kidney disease, progressive inflammation and injury to the renal interstitium are secondary events following glomerular or vascular injury. Tubule epithelial cells synthesize cytokines and chemokines and accumulate complement components that recruit inflammatory cells and lymphocytes into the interstitium, causing progressive fibrosis.

RESIDENT MONOCYTES/DENDRITIC CELLS

The interstitium of normal kidneys contains numerous resident monocytic myelocytes,²³¹ which express dendritic cell (DC) markers and can indeed present antigens.²³¹ DCs have been described to form an immune sentinel network through the entire kidney, where they probe the environment in search of antigens.²³² An inflammatory environment converts the tolerogenic status of resident DCs into an immunogenic one, favoring recruitment of T cells. It is known that cross-presentation by DCs is a major mechanism for the immune surveillance of tissue against foreign antigens.²³³

The role of resident DCs that accumulate in the renal parenchyma of non-immune-mediated proteinuric nephropathies remains poorly understood. However, studies have provided new insights into the activation of DC in the setting of proteinuria. Administration of ovalbumin, which is freely filtered by the glomerulus, to normal mice leads to concentration of the protein principally in proximal tubules and to its transfer to DC in the kidney and renal lymph nodes.²³⁴ Here, ovalbumin is presented to CD8⁺ T cells, thereby inducing proliferation of these cells.

The importance of kidney DC activation to renal injury has been demonstrated by the fact that in transgenic NOH mice (that selectively express the model antigens ovalbumin and hen egg lysozyme in podocytes), DC depletion resolved established periglomerular mononuclear infiltrates.²³⁵ In vitro experiments have also shown that exposure of rat proximal tubule cells to excess autologous albumin, as in the case of proteinuric nephropathies, results in the formation of the N-terminal 24-residue fragment of albumin (ALB₁₋₂₄).²³⁶ This peptide is taken up by DC, where it is further processed by proteasomes into antigen peptides. These peptides were shown to have the binding motif for MHC class I and to be capable of activating CD8⁺ T cells. Moreover, in vivo, in the rat 5/6 nephrectomy model, accumulation of DCs in the renal parenchyma peaked 1 week after surgery and decreased thereafter, concomitant with their appearance in the renal draining lymph nodes. DCs from renal lymph nodes loaded with the albumin peptide ALB₁₋₂₄ activated syngeneic CD8⁺ T cells in primary culture.²³⁶ Thus inflammatory stimuli released from damaged tubules after protein overload may represent danger signals that, in the presence of albumin peptides, alert DCs to promote local immunity via CD8⁺ T cells that are activated in regional lymph nodes and recruited in the renal interstitium.

MACROPHAGES AND LYMPHOCYTES

The interstitial infiltrate of most human chronic renal diseases consists of a number of different effector cells including macrophages, CD4⁺ T cells, and CD8⁺ T cells.²³⁷ In animal models, macrophages are the dominant infiltrating cells in both the early and later stages of chronic renal injury. More specifically, tubulointerstitial macrophage accumulation in chronic nephropathies correlates with the severity of the glomerular and interstitial lesions and degree of renal dysfunction.²³⁷ Recent evidence has elucidated how inflammatory signals from tubule epithelial cells initiate interstitial infiltration of macrophage. In vitro and in vivo experimental studies in 5/6 nephrectomy rat models have documented that albumin triggers tubular epithelial cells to release exosomes loaded with the inflammatory cytokine CCL2 mRNA. The exosomes are delivered to interstitial macrophages, inducing their activation and autocrine recruitment of additional myeloid cells.²³⁸ Direct damage to resident cells is caused through the generation by macrophages of reactive oxygen species (ROS), nitric oxide (NO), complement factors, and proinflammatory cytokines.²³⁹ Macrophages can also affect the supporting matrix and vasculature through expression of metalloproteinases and vasoactive peptides.

Macrophages are only one component of the cellular infiltrate that characterizes inflammation in the renal interstitium. Models of overload proteinuria have emphasized the importance of tubulointerstitial infiltration with mononuclear cells. Indeed, T-helper cells and cytotoxic T cells, as well as macrophages, are observed in the tubulointerstitial infiltrate 2 weeks after protein overload.²⁴⁰ T-cell depletion with intraperitoneal anti-T-cell monoclonal antibody administration did not modify macrophage infiltration, indicating that the influx of these cells was independent of lymphocytes²⁴⁰ and more likely resulted from local tubule cell expression of osteopontin, MCP-1, and the adhesion molecules VCAM and ICAM.

T lymphocytes are also abundantly present in the tubulointerstitial infiltrate early after renal mass ablation in rats and remain there in significant numbers for the following weeks.²⁴¹ While the infiltration of macrophages is part of a nonspecific inflammatory reaction, the presence of lymphocytes within lesions indicates that their recruitment and activation are mediated by an antigen-specific immune response. Their role is to maintain and amplify the inflammatory response in the renal interstitium.

Because B cells are considered to be important mostly in lymph nodes, spleen, and in humoral immune responses, little attention has been paid to their potential role as intrarenal infiltrating cells.²⁴² However, a prominent accumulation of CD20⁺ B cells has been described in membranous nephropathy.²⁴³ Furthermore, CD20⁺ B cells formed a prominent part of the infiltrating cells in renal biopsies from patients with IgA nephropathy and chronic interstitial nephritis.²⁴⁴ Together with CD3⁺ T cells, the CD20⁺ B cells formed large nodular structures, like tertiary lymphatic organs in inflamed tissues.²⁴⁵

In the interstitium, B cells may release proinflammatory cytokines and chemokines, present antigens, and activate T cells, as well as play a role in the development of tissue fibrosis.²⁴⁵

BONE MARROW-DERIVED FIBROCYTES

In proteinuric renal disease, chemokines generated in the inflammatory milieu may contribute to the recruitment of bone marrow–derived fibrocytes to the renal interstitium.²⁴⁶ Fibrocytes are circulating connective tissue cell progenitors with a high capacity for collagen I synthesis. In progressive kidney fibrosis induced by unilateral ureteral obstruction in mice, fibrocytes infiltrated the interstitium, and the number of these cells increased with the progression of fibrosis.²⁴⁷ In addition, the number of infiltrating fibrocytes correlates well with the extent of interstitial fibrosis in several human kidney diseases.²⁴⁶ Although fibrocytes isolated from mice and humans express chemokine receptors including CCR2, CCR3, CCR5, CCR7, and CXCR4,²⁴⁶ the specific chemokine and receptor pair involved in the recruitment of these cells in the damaged tubulointerstitium remains uncertain.

FIBROBLASTS: ACTIVATION AND DEPOSITION OF EXTRACELLULAR MATRIX

The process of tubulointerstitial fibrosis involves the loss of renal tubules and accumulation of myofibroblasts and extracellular matrix (ECM) proteins.²⁴⁸ Resident interstitial fibroblasts and myofibroblasts proliferate in response to macrophage-derived profibrogenic cytokines, and their number correlates with the subsequent formation of a scar.²⁴⁹ Myofibroblasts may be derived from several cellular events including tubular epithelial-mesenchymal transition (EMT),²⁵⁰ endothelial-mesenchymal transition (EndMT),²⁵¹ resident renal fibroblast and pericyte activation and differentiation,²⁵² bone marrow–derived fibrocytes,²⁵³ and recently recognized macrophage-myofibroblast transition (MMT).²⁵⁴

EMT has been suggested as a process that contributes to interstitial fibrosis in CKDs by transformation of injured renal tubule cells into mesenchymal cells.²⁵⁵ However, the evidence for EMT in the adult kidneys and chronic renal disease is controversial and no solid data support EMT as an *in vivo* process in kidney fibrosis.^{256–258}

Activated renal fibroblasts may secrete chemokines, which, in turn, may further attract macrophages and perpetuate tubulointerstitial injury.²¹⁸ Eventually activated fibroblasts produce interstitial matrix components, contributing to interstitial collagen deposition and fibrosis. Increased tubulointerstitial fibrosis is a common feature of kidney injury and results from accumulation of ECM structural proteins. Although it is known that the ECM network undergoes dynamic changes in composition and contents during the progression of CKD, the mechanisms responsible for the initiation and progression of renal fibrosis are not completely understood.²⁵⁹ A recent study using decellularized kidney tissue scaffold from mice with CKD and proteomic profiling has characterized ECM proteins in normal and fibrotic kidneys.²⁶⁰ Among a total of 172 differentially expressed proteins identified in scaffold from diseased kidneys, a core set of 9 signature proteins has been identified, which could be instrumental in establishing an oxidatively stressed, proinflammatory, profibrotic, and antiangiogenetic environment. Among these 9 proteins, glutathione peroxidase 3 (GPX3), an extracellular antioxidant enzyme, was the only protein with downregulated expression in CKD mice. Knockdown of

GPX3 *in vivo* created an oxidatively stressed setting, which increased ECM expression and promoted kidney fibrosis via a NOX2/ROS/p38 MAPK pathway in a model of ureteral obstruction injury.

ECM is maintained by continuous remodeling through the proteolytic action of matrix metalloproteinases (MMPs) and synthesis of new proteins.²⁶¹

MMPs are inhibited by tissue inhibitors of matrix metalloproteinases (TIMPs). Therefore the balance between TIMPs and MMPs determines the ECM integrity. Among the members of the TIMP family, TIMP3 is unique in that it is ECM bound and highly expressed in the kidney.²⁶² TIMP3^{−/−} mice had more interstitial fibrosis, increased synthesis and deposition of type I collagen, increased activation of fibroblasts, and greater activation of MMP2 after unilateral obstruction than wild-type mice.²⁶³ TIMP3 levels are upregulated in patients with diabetic and chronic allograft nephropathy.²⁶³

Recent studies link fibrosis to changes in miRNAs.^{108,264,265} A number of miRNAs have been shown to be relevant to fibrotic processes in diabetic nephropathy including miR-29 and miR-200 families, miR-192, and miR-21.^{265–268} These miRNAs are regulated by TGF- β in renal cells, and normalization of their expression ameliorates fibrosis in *in vitro* and *in vivo* models of diabetes.²⁶⁷

More recently, miR-184 has been shown to be a downstream effector of albuminuria driving renal fibrosis in rats with diabetic nephropathy.²⁶⁹ Indeed, in ZDF rats, miR-184 showed the strongest differential upregulation compared with lean rats (18-fold). Tubular localization of miR-184 was associated with reduced expression of lipid phosphate phosphatase 3 (LPP3) and collagen accumulation. In ZDF rats, ACE inhibitor treatment limited albuminuria and reduced miR-184, with tubular LPP3 preservation and tubulointerstitial fibrosis amelioration. Albumin-induced miR-184 expression in tubule cells was epigenetically regulated through DNA demethylation and histone lysine acetylation and was accompanied by binding to NF- κ B p65 subunit to miR-184 promoter. These results suggest that miR-184 may act as a downstream effector of albuminuria through LPP3 to promote tubulointerstitial fibrosis and offer the rationale to investigate whether targeting miR-184 in association with albuminuria-lowering drugs may be a new strategy to achieve fully antifibrotic effects, at least in diabetic nephropathy.²⁶⁹ In proteinuric CKD associated with type 2 diabetes, an accelerated senescence phenotype has been observed mainly in tubular cells.²⁷⁰ Cellular senescence is an irreversible fate of stress-induced damaged cells that features growth arrest and DNA damage.²⁷¹ Expression of Wnt/ β catenin signaling, an evolutionary pathway involved in tissue repair,²⁷² and particularly Wnt9a is induced in kidney biopsies of patients with proteinuric nephropathies including IgA, diabetic, and membranous nephropathy.²⁷³ *In vitro* evidence indicates that Wnt9a markedly upregulated senescence proteins in proximal tubular cells associated with increased TGF- β 1 production, which eventually promoted proliferation and activation of kidney fibroblasts.²⁷³ Notably, TGF- β 1 and Wnt signaling set up a vicious activation that may promote renal fibrosis, as shown in an ischemia/reperfusion injury mouse model.²⁷³

CHRONIC HYPOXIA

One of the most important contributors to the development of tubulointerstitial fibrosis is chronic ischemia.²⁷⁴ Production of angiotensin II and inhibition of production of nitric oxide underlie chronic vasoconstriction, which may contribute to tissue ischemia and hypoxia.²⁷⁵ In that regard, histologic studies on biopsies from animal models and human kidneys have documented that there is often a loss of peritubular capillaries in areas of tubulointerstitial fibrosis.²⁵⁵ Downregulation of VEGF may be functionally implicated in the progressive attrition of peritubular capillaries and tissue hypoxia, as shown in mouse folic acid nephropathy.²⁷⁶

Pericytes play a critical role in the stabilization and proliferation of peritubular capillaries via interaction with endothelial cells.^{277,278} This process is mediated by several angioregulatory factors including angiopoietin-1, produced by pericytes, and angiopoietin-2, produced by activated endothelial cells.^{278,279} Renal ischemia, as it occurs in CKD due to microvascular rarefaction,²⁸⁰ promotes an imbalance in angiopoietins that, besides leading to proliferation of pericytes, may induce interstitial fibrosis in the long term.²⁸¹

Moreover, given that the size of the interstitial compartment determines the diffusion distance between peritubular capillaries and tubule cells, interstitial fibrosis further impairs tubular oxygen supply. Focal reduction of capillary blood flow leading to starvation of tubules may underlie tubular atrophy and loss. Under these conditions, the remaining tubules are subjected to functional hypermetabolism with increased oxygen consumption, which in turn creates an even more severely hypoxic environment in the renal interstitium. In vitro, such hypoxia stimulates fibroblast proliferation and ECM production by tubular epithelial cells.²⁸²

ENDOGENOUS SYSTEMS OF TISSUE REPAIR

PROTECTIVE MACROPHAGES

Much remains to be learned about macrophages in tubulointerstitial injury. The role of interstitial macrophages was elucidated in mice with progressive Adriamycin-induced nephropathy.²⁸³ By treating mice with the monoclonal antibody ED7 directed against the CD11b/CD18 integrin, which is expressed by macrophages, renal cortical macrophages (ED1 positive cells) were reduced by almost 50%, whether ED7 was administered before or after Adriamycin administration.²⁸⁴ However, ED7 reduced renal structural and functional injury only when treatment was started before Adriamycin administration.²⁸⁴

Among several possible explanations for these observations is a temporal change in the predominant macrophage phenotype. If pathogenic macrophages predominated early and protective macrophages later in the course of the disease, then only early antimacrophage treatment would be expected to protect against progression. Indeed, macrophages can exhibit distinctly different functional phenotypes and can be polarized toward a proinflammatory (M1 macrophages) or tissue-reparative (M2 macrophages) phenotype.²⁸⁵ In the peritubular interstitium, macrophages have been shown to mediate tissue repair in response to acute kidney injury by adopting an M2 phenotype and producing a cytokine environment that supports tubular

repair and proliferation rather than inflammation.²⁸⁶ Colony-stimulating factor-1 (CSF-1) signaling mediated M2 macrophage-induced recovery from renal injury, since pharmacologic blockade of CSF-1 decreased M2 polarization and eventually inhibited tissue repair.²⁸⁷ The subtypes of M2 macrophages (M2a, M2b, and M2c)²⁵⁴ are thought to suppress immune responses and promote tissue repair, but with different and sometimes controversial functions.²⁸⁵

Studies have also demonstrated marked macrophage heterogeneity and context specificity, depending on the nature of the injury and location within the kidney.²⁸⁸ Evidence is available that macrophages perform both injury-inducing and repair-promoting tasks in different models of inflammation. This has been shown in a reversible model of liver injury, in which the injury and recovery phases are distinct.²⁸⁹ Macrophage depletion when liver fibrosis was advanced resulted in reduced scarring and fewer myofibroblasts. Macrophage depletion during recovery, by contrast, led to a failure of matrix degradation.²⁸⁹ These findings provide clear evidence that a functionally distinct subpopulation of macrophages exists in the same tissue.

Further studies on possible temporal variations in the phenotype, activation status, and net effect on injury of macrophages should give a better understanding of the complex role of macrophages in tubulointerstitial injury and repair of chronic renal disease, particularly in the proteinuric setting.

REGULATORY T CELLS

CD4⁺ T cells constitute a critical component of the adaptive immune system and are typified by their capacity to help both humoral and cell-mediated responses. However, there is a substantial functional diversity among CD4⁺ T cells and, clearly, certain subpopulations hinder rather than help immune response. The most well-characterized example of an inhibitory subpopulation is CD4⁺CD25⁺, which appears to play an active role in downregulating pathogenic autoimmune responses.²⁹⁰ CD4⁺CD25⁺ T cells are potent immunoregulatory cells that suppress T-cell proliferation in vitro and have the capacity to suppress immune responses to autoantigens and alloantigens, tumor antigens, and infectious antigens in vivo.²⁹¹

The regulatory activity of these cells in the setting of chronic renal diseases is highlighted by studies in SCID mice reconstituted with CD4⁺CD25⁺ T cells after induction of Adriamycin nephrosis.²⁸⁴ Mice reconstituted with these regulatory cells had significantly reduced glomerulosclerosis, tubular injury, and interstitial expansion compared with unreconstituted mice with Adriamycin-induced nephrosis.

A study using the green fluorescence protein (GFP)-*Foxp3* mouse suggests that *Foxp3* expression identifies the regulatory T-cell (Treg) population.²⁹² In the murine model of Adriamycin nephropathy, the adoptive transfer of *Foxp3*-transduced T cells protected against renal injury. Urinary protein excretion and serum creatinine were reduced, and there was significantly less glomerulosclerosis, tubular damage, and interstitial infiltrates.²⁹³ More recent evidence in a mouse model of glomerulonephritis has shown a long-term amplification of kidney Treg cells post injury.²⁹⁴ Treg depletion 21 days post disease induction resulted in further impairment of renal function with a concomitant increase in Th1 cells.²⁹⁴ Kidney infiltrating Treg cells displayed a

molecular profile resembling that of other non-lymphoid tissue Tregs, namely the expression of GATA3, the IL33 receptor subunit ST2, PPAR γ , and the chemokine receptor CCR4.²⁹⁴

KIDNEY-DERIVED PROGENITOR CELLS

In chronic proteinuric renal disease, regression of glomerular structural changes is associated with remodeling of the glomerular architecture.²⁹⁵ Instrumental to this discovery were three-dimensional reconstruction studies of the glomerular capillary tuft, which allowed the quantification of sclerosis volume reduction and capillary regeneration upon treatment.²⁹⁵ The reversal of early glomerular damage in animal models and humans²⁹⁶ argues for the existence of a regenerative mechanism that promotes glomerular repair. However, mature podocytes are postmitotic cells with limited capacity to divide in situ and therefore unable to regenerate.²⁹⁶ A potential mechanism for podocyte replacement by bone marrow-derived stem cells has been described in the Alport mouse model, as well as in kidney transplants.^{297,298} Nevertheless, most studies concluded that regeneration occurs predominantly from resident renal progenitors,^{54,65} although the source of these cells remains ill-defined.

Recently, a study using a triple-transgenic mouse model that allowed permanent marking of glomerular parietal cells and their progeny upon administration of doxycycline showed that PECs of the Bowman capsule possess the capability to migrate into the glomerular tuft via the vascular stalk, where they differentiate into podocytes.²⁹⁹ Similarly, in the adult human kidney, cells localized between the urinary pole and vascular pole of the Bowman capsule, which expressed both progenitor and podocyte markers (CD24⁺CD133⁺PDX⁺ cells), can differentiate into podocytes by losing stem cell markers and expressing markers indicative of a podocyte phenotype while progressing from the urinary pole to the surface of the glomerular tuft.³⁰⁰

More intriguing from the clinical perspective is the finding that ACE inhibition induces glomerular repair in Munich Wistar Fromter rats, a model of spontaneous glomerular injury.³⁰¹ In these proteinuric animals, besides halting age-related podocyte loss, lisinopril increased the number of glomerular podocytes above baseline, which was associated with an increased number of proliferating Wilms tumor (WT)-1-positive cells, loss of cycling-dependent kinase inhibitor p27 expression, and increased number of parietal podocytes. This indicates that remodeling of Bowman capsule epithelial cells contributes to the ACE inhibitor-induced restructuring of the damaged glomerular capillary, primarily by restoring the podocyte population.

Similarly, glomerular repair is augmented when glucocorticoid treatment is given to mice with experimental FSGS at a time when podocyte number was already decreased.³⁰² Prednisone increased podocyte number, which correlated with reduced proteinuria and decreased glomerulosclerosis.

Beside ACE inhibitors^{158,165} and prednisone,³⁰² Notch inhibitors,³⁰³ blockers of chemokine stromal-derived factor-1,³⁰⁴ and retinoids³⁰⁵ can be added to the list of agents that improve podocyte regeneration by augmenting the number of PECs progenitors. Indeed, in vitro exposure of human renal progenitor cells to human serum

albumin inhibited their differentiation into podocytes by sequestering retinoic acid and preventing retinoic-acid-response element-mediated transcription of podocyte-specific genes.¹⁵⁹ Similarly, in vivo in mice with Adriamycin-induced nephropathy, a model of human FSGS, blocking endogenous retinoic acid synthesis increased proteinuria and exacerbated glomerulosclerosis.¹⁵⁹ This effect was related to a reduction in podocyte number. Together, these experimental studies suggest that restoring the capacity of parietal epithelial progenitor cells to differentiate into podocytes could promote the regeneration of podocytes and potentially result in the regression of glomerular disease.

SYSTEMIC CONSEQUENCES OF NEPHROTIC-RANGE PROTEINURIA

Nephrotic-range proteinuria is accompanied by a cluster of abnormalities that is known collectively as the nephrotic syndrome. It is characterized by systemic complications that result from profound alterations in the composition of the body protein pool, a state of sodium retention, dyslipidemia, abnormalities of coagulation factors, and a variable degree of renal insufficiency.

HYPOALBUMINEMIA

Clinical manifestations of nephrotic syndrome become evident in patients with levels of proteinuria exceeding 3.5 g/day. Proteinuria in overtly nephrotic subjects usually exceeds this lower bound by a factor of 2 to 3. Immunochemical analysis shows that albumin accounts for >80% of the excreted proteins.³⁰⁶ The second most copiously excreted protein is immunoglobulin, which after albumin is the next most abundant protein in plasma. One of the most common systemic abnormalities associated with nephrotic proteinuria is hypoalbuminemia, which develops in most patients.

PATHOGENESIS OF HYPOALBUMINEMIA

Under normal conditions, albumin production by the liver is 12 to 14 g/day (130–200 mg/kg). Production equals the amount catabolized, predominantly in extrarenal locations.³⁰⁷ However, about 10% is catabolized in the proximal tubule of the kidney after reabsorption of filtered albumin.³⁰⁷ In patients with nephrotic syndrome, hypoalbuminemia results from excessive urinary loss, decreased hepatic synthesis, and increased rates of albumin catabolism (Fig. 29.9).

Urinary albumin loss is an important contributor to the development of hypoalbuminemia. However, it is not a sufficient cause in most patients with nephrotic syndrome because the rate of hepatic albumin synthesis can increase by at least threefold, thereby compensating for urinary albumin loss.³⁰⁷ Enhanced loss of albumin in the gastrointestinal tract has also been proposed to contribute to hypoalbuminemia, but there is little evidence for this hypothesis.³⁰⁸ Therefore for hypoalbuminemia to develop, there must be either an insufficient increase in hepatic synthetic rate or increase in albumin catabolism.

Normally the rate of hepatic albumin synthesis may increase by as much as 300%. However, studies of nephrotic syndrome in animal models and in humans with

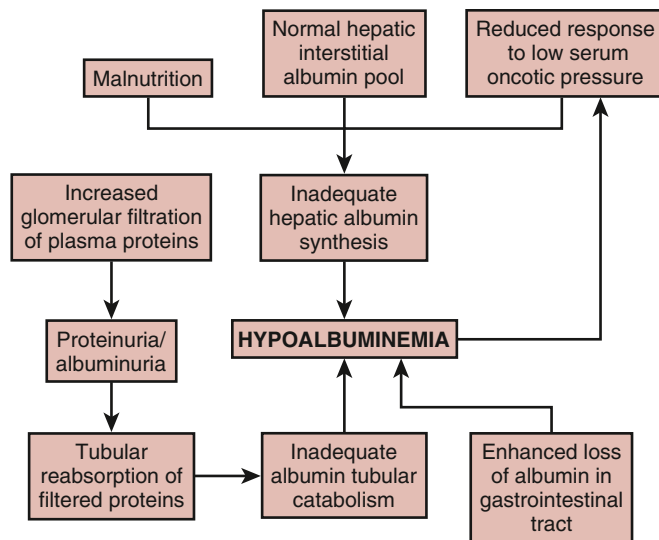


Fig. 29.9 Schematic representation of mechanisms leading to nephrotic hypoalbuminemia. Compensatory mechanisms, such as increase in albumin synthesis and decrease in albumin catabolism, are insufficient to correct the hypoalbuminemia.

hypoalbuminemia demonstrate that the rate of albumin synthesis is at or only slightly above the upper limit of normal as long as dietary protein is adequate.³⁰⁹ This indicates an inadequate synthetic response to hypoalbuminemia by the liver.

Oncotic pressure of the plasma perfusing the liver is one major regulator of protein synthesis.³¹⁰ Experimental evidence in rats that are genetically deficient in circulating albumin showed a twofold increase in the hepatic transcription rate of the albumin gene compared with normal rats.³¹⁰ However, in these rats the increase in hepatic albumin synthesis was inadequate to compensate for the degree of hypoalbuminemia, which indicates an impaired synthetic response.³¹⁰ Similarly, in nephrotic patients, reduced oncotic pressure is unable to enhance the albumin synthetic rate of the liver to the extent required to restore plasma albumin concentration.³⁰⁹

The contribution of renal albumin catabolism to hypoalbuminemia in nephrotic syndrome is controversial. Some have argued that the renal tubular albumin transport capacity is already saturated at physiologic levels of filtered albumin and that any increase in filtered protein, instead of being absorbed and catabolized, is simply excreted in the urine.³¹¹ Studies in isolated perfused proximal tubules in rabbits, however, demonstrated a dual transport system for albumin uptake.³¹² In addition to a low-capacity system that became saturated when the protein load exceeded physiologic levels, a high-capacity low-affinity system was also present and allowed the tubular absorptive rate for albumin to increase as the filtered load rose. Thus an increase in the fractional catabolic rate may occur in nephrotic syndrome.

This hypothesis is supported by the positive correlation between fractional albumin catabolism and albuminuria in rats with puromycin aminonucleoside-induced nephrosis.³¹³ Nevertheless, since total body albumin stores are substantially decreased in nephrotic syndrome, absolute catabolic rates may be normal or even reduced.³⁰⁸

In summary, hypoalbuminemia in nephrotic syndrome results from multiple alterations in albumin homeostasis that are not sufficiently compensated for by hepatic albumin synthesis and by decreased renal tubular albumin catabolism.

CONSEQUENCES OF HYPOALBUMINEMIA

Impairment of kidney function is the rule in patients with nephrotic hypoalbuminemia and usually manifests in two ways. One is the inability of the kidney to maintain sodium and fluid homeostasis. The other is loss of intrinsic ultrafiltration capacity of glomerular capillary walls, a phenomenon that leads, in turn, to fall in GFR.³¹⁴

By estimating GFR and its determinants in humans, it has been shown that a reduced GFR in some forms of nephrotic syndrome (minimal-change and membranous nephropathy) is exclusively a consequence of profoundly lowered hydraulic permeability.³¹⁵ In nephrotic syndrome associated with lupus nephritis, idiopathic focal and segmental glomerulosclerosis, and diabetic nephropathy, both reduction of the surface area available for filtration and impaired hydraulic permeability contribute to depression of the ultrafiltration coefficient (Kf).^{316,317}

The principal cause of impaired hydraulic permeability in nephrotic disorders is broadening and effacement of epithelial foot processes.³¹⁵ This lowers the frequency of interpodocytic slit diaphragms through which water must pass to gain access to the Bowman space, thereby increasing resistance to water flow. The low Kf is partially offset by an increase in net ultrafiltration pressure, which is largely due to a substantial lowering of the intraglomerular capillary oncotic pressure. As a result, the fall in GFR is not proportional to the decrease in Kf. This compensatory elevation in net ultrafiltration pressure explains why reduced values of single nephron GFR are not consistently observed in all experimental nephrotic models.³¹⁸

The low ultrafiltration capacity induced by glomerular disease and protein depletion makes the nephrotic patient particularly vulnerable to acute exacerbations of hypofiltration and renal insufficiency.³¹⁹ As the prevailing level of GFR depends heavily on ultrafiltration pressure in the presence of a low Kf, any maneuver lowering the glomerular capillary perfusion pressure can therefore cause a precipitous fall in the GFR.

An additional consequence of hypoalbuminemia is the potential for enhanced drug toxicity.³²⁰ Indeed, many drugs are bound to albumin. Hypoalbuminemia reduces the number of available binding sites and increases the proportion of circulating free drug, but in the steady state this is counterbalanced by faster metabolism. Furthermore, since protein binding may enhance tubule drug secretion, diminished protein binding in nephrotic syndrome may delay renal excretion of some drugs.³²¹ Although the clinical consequences of altered protein binding may be difficult to predict, higher levels of free drug may be toxic, as shown with prednisolone.³²²

The case of diuretics is intriguing. Resistance to loop diuretics, which often occurs in patients with nephrotic syndrome, may be due to reduced delivery of the diuretic to its site of action, secondary to hypoalbuminemia. Anecdotal reports suggest that the administration of furosemide with small amounts of albumin (6–20 g) can enhance the response

to furosemide in nephrotic patients.³²³ These observations are not conclusive because others did not show a difference in excretion of intravenous furosemide in the urine of nephrotic patients compared with normal controls.³²⁴ On the other hand, excessive amounts of filtered albumin in the tubule may bind furosemide and make it less effective.³²⁵

Many binding proteins are lost in the urine in nephrotic syndrome.³²⁶ Consequently, in patients with nephrotic syndrome, the plasma levels of many ions (iron, copper, and zinc), vitamins (vitamin D metabolites), and hormones (thyroid and steroid hormones) are low because the level of protein-bound ligands is reduced. Urinary loss of protein-bound ligands can theoretically cause depletion, but there is little convincing clinical evidence for this, with the possible exception of vitamin D.^{327,328}

Although the hypocalcemia of nephrotic syndrome was once attributed solely to the reduction in protein-bound calcium secondary to hypoalbuminemia, a subset of patients has been noted with hypocalcemia out of proportion to the hypoalbuminemia. In these patients ionized serum calcium is decreased.³²⁹ Secondary hyperparathyroidism is seen in some patients, even in the absence of renal failure, as are changes in bone histology consistent with mixed osteomalacia and osteitis fibrosa cystica bone disease.³³⁰ Not all investigators, however, have observed abnormalities in calcium homeostasis in nephrotic syndrome.³³¹

Finally, hypoalbuminemia may play a role in platelet hyperaggregability.³³² Because albumin normally binds arachidonic acid, thus limiting its conversion to thromboxane A_2 by platelets, hypoalbuminemia might allow increased platelet arachidonate metabolism to take place and platelet hyperreactivity may result.³³²

EDEMA FORMATION

The clinical manifestation that most frequently brings the nephrotic patient to medical attention is the formation of edema. This represents an increase in the size of the interstitial fluid compartment. The interstitial fluid accumulates most readily in dependent areas where tissue pressure is low. It thus manifests as periorbital edema upon awakening in the morning and pedal edema at the end of the day. Even when edema is generalized and massive, a condition referred to as anasarca, it remains most marked in the lower extremities. The mechanisms responsible for extravascular fluid accumulation in nephrotic patients are complex and only partially understood.

REDUCED PLASMA ONCOTIC PRESSURE

Low colloid oncotic pressure because of hypoalbuminemia favors movement of water from the intravascular to the interstitial space. Under normal conditions, edema formation is halted by expansion and proliferation of lymphatics that increase lymphatic flow, as well as by reduction of interstitial oncotic pressure due to protein-free fluid accumulation. In addition, the increased hydraulic pressure in the interstitium because of fluid accumulation lowers the transcapillary pressure gradient, further reducing the transudation of plasma fluid into the interstitial space. However, there is no clear evidence of alterations in these normal defense mechanisms against edema formation in nephrotic patients.^{333,334} For example, comparable changes

in interstitial and plasma colloid oncotic pressure have been documented during relapse and remission phases in patients with nephrotic syndrome.^{333,334} These observations suggest that hypoalbuminemia per se may not be the primary determinant of the severity of edema formation and that intrarenal mechanisms may have a prominent contributory role.

ALTERATIONS IN BLOOD VOLUME

According to the traditional view, lowering of the plasma albumin concentration eventually induces renal sodium and fluid retention in nephrotic syndrome by causing hypovolemia, the so-called “underfill mechanism” (Fig. 29.10). Indeed, hypovolemia as a consequence of reduced plasma colloid oncotic pressure triggers a cascade of events that signal the kidney to retain filtered sodium and water.³³⁵ Thus hypovolemia is the afferent stimulus of a complex pathway of responses mediated by low- and high-pressure baroreceptors in the cardiac atria, carotid arteries, and aorta that activate the sympathetic nervous system and renin-angiotensin system. Moreover, hypovolemia also promotes excessive nonosmotic secretion of arginine vasopressin, which further contributes to water retention by the kidneys.³³⁵

The homeostatic response of renal sodium and water retention that serves to restore intravascular volume also exacerbates hypoalbuminemia, thereby sustaining transudation of plasma fluid into the interstitial space. The fact that salt retention may be the consequence of

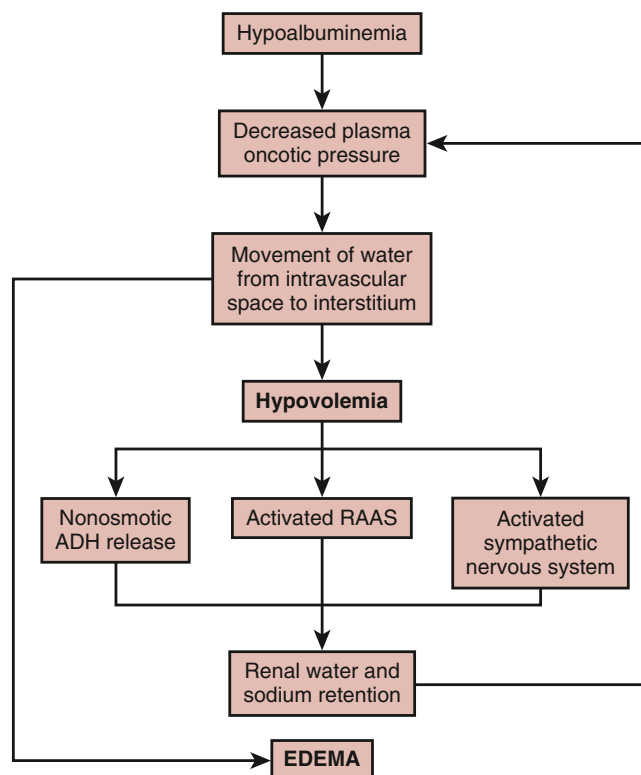


Fig. 29.10 The “underfill” mechanism of edema formation. Hypovolemia, as a consequence of reduced plasma oncotic pressure, is the key event that signals the kidney to retain filtered sodium and water. ADH, antidiuretic hormone; RAAS, renin-angiotensin-aldosterone system.

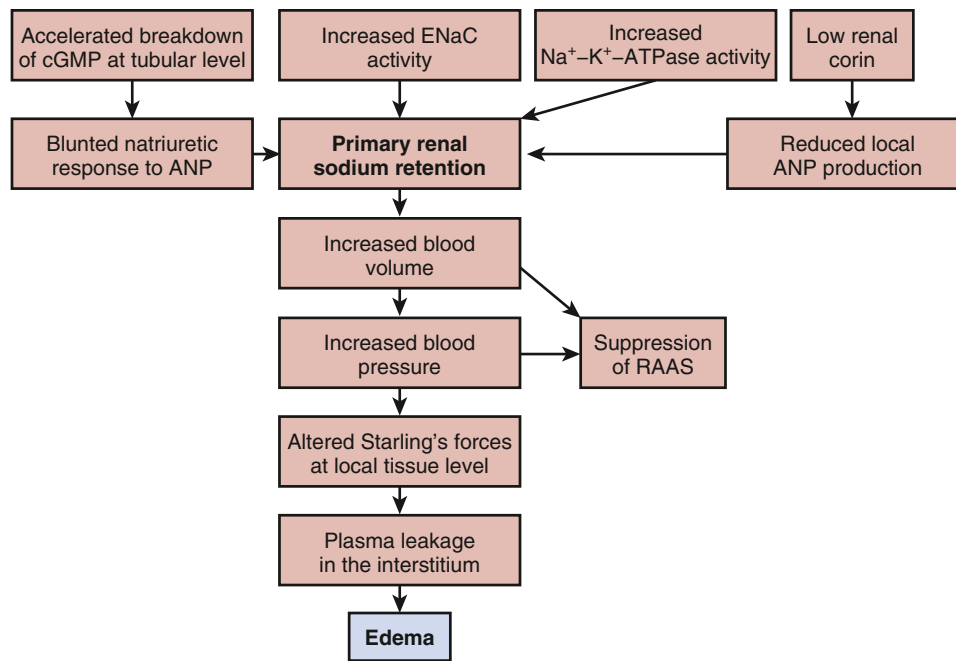


Fig. 29.11 The “overflow” mechanism of edema formation. The abnormal renal sodium retention is the consequence of blunted natriuretic response to ANP, increased epithelial sodium channel (ENaC) activity and Na^+/K^+ ATPase activity and is the key event of the process. The resulting hypervolemia alters Starling’s forces across the capillary wall at local tissue level, leading to overflow edema. cGMP, Cyclic guanosine monophosphate; RAAS, renin-angiotensin-aldosterone system.

an underfilled circulation is consistent with the finding that head-out water immersion, a maneuver that increases plasma volume, is followed by a natriuretic and diuretic response in some nephrotic patients.³³⁶

This mechanistic scenario of edema formation in nephrotic syndrome would also imply consistently reduced plasma volume,³³⁷ as well as an elevated plasma renin activity (PRA),³³⁸ and increased plasma and urinary levels of catecholamines.³³⁹ However, only a minority of nephrotic patients have a low plasma volume³³⁹; in fact, approximately 70% of patients had normal or even high values in some studies.³⁴⁰ In some cases, plasma volume was lower during remission than during the acute phase of the disease.^{339,340} However, methodologic issues have been raised about the measurement of plasma volume in nephrotic patients that may limit the interpretation of these studies.³⁴¹

Measurement of vasoactive hormones, which are responsive to low plasma volume and can be taken as surrogate markers of the intravascular volume, also documented that only 50% of nephrotic patients have higher than normal PRA and plasma, as well as urinary aldosterone levels.³⁴² Moreover, pharmacologic blockade of the renin-angiotensin-aldosterone system in nephrotic patients with a high PRA does not change sodium excretion.³⁴³ Evidence that PRA often increases rather than decreases after steroid-induced remission of nephrotic syndrome is additional, albeit indirect, evidence that argues against a key role for hypovolemia in edema formation in most nephrotic patients.³⁴²

INTRARENAL MECHANISMS

Alternatively, the “overflow” theory hypothesizes that there is a dominant mechanism by which the kidneys retain sodium independently of circulating plasma volume,

leading to hypervolemia (Fig. 29.11).³³⁵ Examination of the edema-forming, nephrotic patient during consumption of a known amount of sodium reveals a positive sodium balance. This results in increased blood volume, which by altering Starling’s forces across the capillary wall leads to plasma leakage into the interstitium and overflow edema. This mechanism has been illustrated in an unilateral model of puromycin aminonucleoside-induced nephrosis in a rat.³⁴⁴ In such a model, in which albumin concentration in the systemic circulation is normal, only the proteinuric kidney (not the contralateral intact one) retained excessive amounts of sodium and water. This indicates that abnormal sodium retention by the proteinuric kidney is brought about by intrarenal rather than circulating or systemic factors.

These findings can be partly explained by a lowered filtered sodium load, a consequence of the diminished GFR that frequently accompanies nephrotic range proteinuria. However, since the fractional sodium excretion is low, enhanced tubular sodium reabsorption appears to be the predominant cause of sodium retention in nephrotic syndrome. Analysis of segmental sodium transport in nephrotic rats has identified the collecting duct as the major site of enhanced sodium reabsorption.³⁴⁴

A crucial observation was that natriuretic and diuretic responses to intravenously infused atrial extract (from normal or nephrotic rats) or synthetic atrial natriuretic peptide (ANP) were markedly reduced in nephrotic as compared with normal rats.³⁴⁵ In a rat model of unilateral glomerulopathy, the blunted natriuretic and diuretic response to ANP was confined to the “nephrotic” kidney as opposed to the contralateral normal kidney, despite a comparable increase in GFR.³⁴⁶ Moreover, both enhanced release of endogenous ANP during water immersion and

infusion of exogenous ANP failed to promote an appropriate natriuretic response in nephrotic patients.³⁴⁷

Taken together, these findings support a role for ANP in intrarenal sodium retention in nephrotic syndrome. In addition, it can be inferred that alterations in the intrinsic transport properties of the collecting duct render this tubule segment unresponsive to the natriuretic action of ANP.

In some studies, increased activity of efferent sympathetic nerve has been related to the blunted ANP natriuretic response.³⁴⁸ More consistent evidence indicates enhanced phosphodiesterase activity in collecting duct cells from nephrotic animals, leading to accelerated breakdown of normally produced cyclic guanosine monophosphate, which is important for intracellular signaling after ANP binding to its specific receptors.³⁴⁹

With the discovery of corin, a 1042 amino acid transmembrane serine protease that converts proANP and proBNP into the active forms ANP and BNP,^{350,351} the pathogenesis of edema formation in nephrotic syndrome has been revised. Induction of nephrotic syndrome by puromycin aminonucleoside or glomerulonephritis by anti-Thy1 induced concomitant increase in proANP and decrease in ANP in the kidney in association with low renal immunoreactive levels of corin.^{352,353} Upregulation of phosphodiesterase 5 (PDE5) and kinase G II secondary to reduced levels of corin resulted in reduced cyclic guanosine monophosphate in the collecting duct and subsequently in increased abundance of the low-conductance epithelial sodium channel (ENaC) seen in nephrotic syndrome and glomerulonephritis.³⁵³ These findings suggest that corin deficiency by lowering locally produced ANP might be involved in primary salt retention seen in edematous glomerular diseases.³⁵⁴ In this regard, reduced urinary corin levels were reported in patients with CKD.³⁵⁵

In the kidney, the ultimate regulation of sodium reabsorption occurs in the late distal convoluted tubule, connecting segment, as well as the collecting duct through the low-conductance ENaC,³⁵⁶ located on the apical membrane of tubular cells. In recent years, from *in vitro* and *in vivo* data, endoluminal activation of ENaC by aberrantly filtered serine proteases has been proposed as an underlying mechanism of sodium retention in nephrotic syndrome.^{357–359} Evidence is available that the proteolytic removal of an inhibitory domain from the γ -subunit of ENaC by the serine protease plasmin can activate ENaC.³⁶⁰ Plasmin, present in the urine of nephrotic rats and humans, has been shown to activate ENaC via this mechanism.³⁶¹ In pediatric patients, acute nephrotic syndrome with volume expansion was associated with significantly more plasmin in the urine compared with the remission phase.³⁶² Additionally, urokinase-type plasminogen activator present in the rat and human kidney can convert inactive plasminogen (which is filtered by the nephrotic kidney) to the active form plasmin.³⁶¹ In the rat puromycin aminonucleoside nephrosis model, amiloride increased urine sodium excretion and reduced ascites volume. This effect was attributed to the ability of amiloride to inhibit both ENaC and urokinase-type plasminogen activator and thus reduce the amount of active plasmin present.³⁶¹ Although plasmin appears to be the dominant active serine protease in nephrotic urine, other urinary proteases including cathepsin B³⁶³ and plasma kallikrein³⁶⁴ are also present. Thus at present, several serine proteases

might contribute redundantly to ENaC activation and sodium retention during nephrotic syndrome, since direct inhibition of specific proteases only partially attenuates proteinuria-induced sodium retention.

ENaC is also regulated by aldosterone.³⁵⁶ In rat models of nephrotic syndrome, activation of ENaC together with elevated plasma aldosterone levels has been reported.³⁶⁵ Nevertheless, in puromycin-induced nephrosis in rats with clamped aldosterone plasma levels, sodium retention persisted even when ENaC recruitment to the apical membrane was inhibited.³⁶⁶ Conversely, the transport activity of sodium-potassium-adenosine triphosphatase (Na^+ , K^+ -ATPase), the ubiquitous sodium pump localized exclusively on the basolateral membrane, was increased.³⁶⁶ These findings indicate that increased Na^+ , K^+ -ATPase activity is the driving force behind enhanced sodium reabsorption in nephrotic syndrome, an observation confirmed by several studies in the cortical collecting ducts in nephrotic rats.³⁶⁷ Because the Na^+ , K^+ -ATPase pump in the basolateral membrane promotes secondary passive sodium entry from the lumen through the ENaC, Na retention in the collecting duct of nephrotic rats can result from the coordinated overactivity of these tubular sodium transporters.

An additional hypothesis concerning intrarenal mechanisms of nephrotic edema proposes that interstitial inflammation of the kidney plays a major role in the pathogenesis of primary sodium retention.³⁶⁸ The generation of vasoconstrictor and reduction of vasodilator substances in the interstitium, driven by the inflammatory cell infiltrate, can lead to a reduction in Kf and single-nephron GFR. These glomerular hemodynamic changes that reduce filtered sodium load combine with the increased net tubular sodium reabsorption induced by mediators released from the inflammatory cell infiltrate leading to primary sodium retention, an “overfilled” intravascular volume, and increased capillary hydrostatic pressure. The decrease in plasma oncotic pressure again promotes fluid movement out of the vascular compartment, thereby buffering the changes in blood volume induced by primary sodium retention. A renal inflammatory infiltrate is, however, minimal or absent in most children with minimal-change nephrotic syndrome. Thus nephrotic edema may derive from a combination of primary sodium retention and relative arterial underfilling. The predominance of one or the other mechanism is perhaps in accordance with the pathogenesis of nephrotic syndrome or the stage of the disease.

Deranged renal water handling is also a cardinal feature of nephrotic syndrome. Defects in both urinary diluting ability³⁶⁹ and concentrating capacity³⁶⁹ have been documented in nephrotic patients. The cause of the concentrating defect has been explored in experimental models of nephrotic syndrome.^{370,371}

HYPERLIPIDEMIA

Both quantitative and qualitative changes in lipid metabolism occur in nephrotic syndrome, with virtually all plasma lipid and lipoprotein fractions being elevated.^{372,373} Blood levels of cholesterol are almost always increased and continue to rise as the severity of nephrotic syndrome increases.³⁷² Total cholesterol and cholesterol esters are all increased.³⁷⁴ Levels

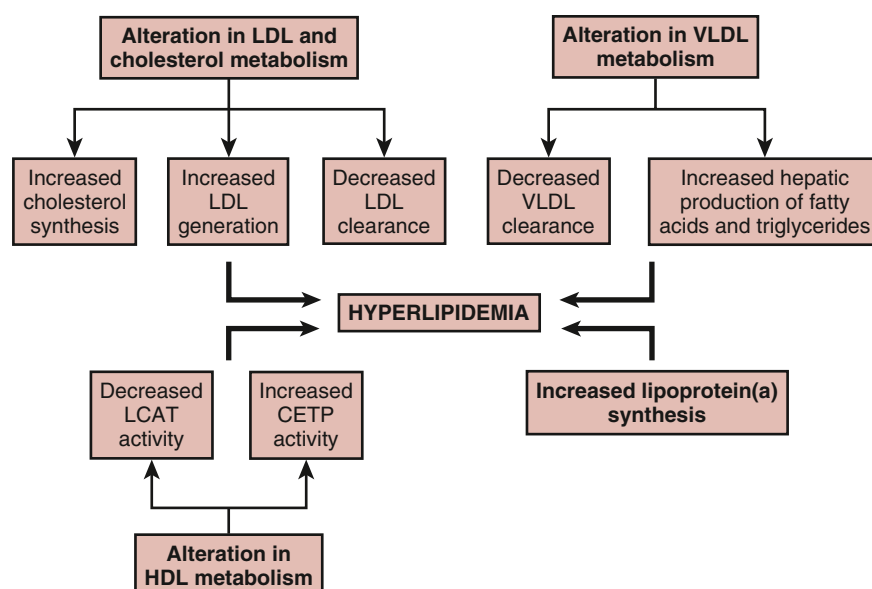


Fig. 29.12 Pathophysiology of nephrotic hyperlipidemia. All abnormalities of lipid profile originate from alteration in low-density lipoprotein (LDL), very-low-density lipoprotein (VLDL), high-density lipoprotein (HDL), and cholesterol metabolism, as well as increased synthesis of lipoprotein(a). *CETP*, cholesteryl ester transfer protein.

of triglycerides are more variable and in many patients do not increase at all, except when the nephrotic state is severe.³⁷² Plasma levels of free fatty acids are within normal limits in nephrotic syndrome, although a smaller than normal fraction is bound to plasma albumin.³⁷² Levels of very-low-density (VLDL), intermediate-density (IDL), and low-density (LDL) lipoproteins increased early in nephrotic syndrome^{372,375}; data on high-density lipoproteins (HDLs) are less clear. Plasma levels are usually normal but may decrease due to HDL excretion in the urine in severely proteinuric patients.³⁷²

PATHOGENESIS OF NEPHROTIC HYPERLIPIDEMIA

Two mechanisms contribute to nephrotic dyslipidemia: overproduction and impaired catabolism/clearance of serum lipids and lipoproteins (Fig. 29.12). There is general agreement that hepatic lipid and apolipoprotein synthesis are both increased and that the clearance of chylomicrons and VLDL³⁷⁶ is reduced in nephrotic syndrome. Cholesterol synthesis has been shown to increase in both animals and humans in response to the hypoalbuminemia associated with nephrotic syndrome.³⁷⁷ Hepatic activity of hydroxymethylglutaryl CoA reductase, the rate-limiting step for hepatic synthesis of cholesterol, is elevated.³⁴⁰ In general, serum cholesterol levels are inversely proportional to serum albumin levels,³⁴⁰ and cholesterol levels generally normalize upon remission. Conversely, triglyceride synthesis does not appear to be increased.

It has been suggested that lipoprotein synthesis increases in parallel with albumin synthesis since they share a common secretory pathway.³⁷⁸ This hypothesis was supported by studies showing that infusion of albumin partially corrected nephrotic hyperlipidemia.

Most evidence still indicates that reduced extracellular albumin concentration or reduced extracellular oncotic pressure, or both, in some way regulate apolipoprotein synthesis and lipogenesis by the liver. Although hepatic apolipoprotein synthesis is increased in nephrotic syndrome,

not all apolipoproteins are affected to the same degree, and mechanisms causing increased synthesis of the various apolipoproteins are also different.³⁷⁹ Secretion of Apo A is increased approximately sixfold,³⁷⁸ whereas synthesis of Apo B and E is only twofold increased.

In addition to increased synthesis, studies in animals and humans have determined alterations in the catabolism of lipids in nephrotic syndrome. The clearance of chylomicrons and VLDL is reduced following the onset of proteinuria but is normal in rats with hereditary analbuminemia,^{378,380} suggesting that urinary loss of a liporegulatory substance and not reduced albumin concentration or oncotic pressure may play a role in causing defective lipolysis.

One possible explanation for the defective removal of lipoproteins is a decrease in the activity of lipoprotein lipase (LPL), which hydrolyzes triglycerides in VLDL and chylomicrons, releasing free fatty acids. LPL activity is reduced in nephrotic rats, which provides a potential mechanism for delayed lipolysis.³⁷⁸ It is now established that the binding of LPL to heparan sulfate proteoglycans on endothelial cells occurs via endothelium-derived glycosylphosphatidylinositol-anchored HDL-binding protein 1 (GPIHBP1),³⁸¹ which is downregulated in patients with nephrotic syndrome.³⁸² Thus a specific reduction in LPL attached to the vascular endothelium may play a role in the reduced catabolism of chylomicrons and VLDL in nephrotic syndrome.^{380–382}

Studies in patients with nephrotic syndrome have not been as detailed as in the rat; however, when comparable studies are evaluated, both species exhibit similar disturbances in lipid metabolism. The fractional turnover rate of triglycerides is reduced in nephrotic subjects compared with controls, and the half-life of triglycerides in VLDL is prolonged from 4 to 11 hours.³⁸³ VLDL catabolism is decreased, and the concentration curve over time has an unusual shape, presumably resulting from a delay in the conversion of VLDL into LDL.³⁸⁴

It has been suggested that the delay in lipolysis in humans, as in rats, is due to a decrease in LPL activity. Evidence supporting this hypothesis is that LPL activity is reduced in children with nephrotic syndrome and increases after remission. Furthermore, there is a strong inverse correlation between LPL activity and the concentration of triglycerides in the VLDL fraction,³⁸⁵ although not all investigators report decreased LPL activity in nephrotic patients.³⁸⁶

LDL catabolism has been shown to be either normal or reduced³⁸⁷ in patients with nephrotic syndrome and only marginally reduced in nephrotic rats.³⁸⁸ Reduced receptor-mediated LDL clearance has been reported in some clinical studies,³⁸⁹ which may account in part for elevations in LDL. In addition, increased expression of proprotein convertase subtilisin/kexin type 9 (PCSK9) results in the increased degradation of the LDL receptor and decreased uptake by the liver.^{390,391} The elevated plasma levels of both cholesterol and PCSK9 detected in patients with active nephrotic syndrome have been reported to return to normal levels upon remission of disease.³⁹²

Nephrotic syndrome is also associated with abnormalities in the activity of enzymes required for effective function of HDL. Cholesterol ester transfer protein (CETP) catalyzes the transfer of the cholesterol ester-rich core of HDL₂ to VLDL remnant particles creating LDL, thereby increasing LDL cholesterol at the expense of HDL cholesterol. CETP is increased in the plasma of nephrotic patients and correlates positively with VLDL cholesterol and negatively with HDL cholesterol.³⁹³

The enzyme lecithin-cholesterol acyl transferase (LCAT) catalyzes the esterification of cholesterol and its incorporation in HDL particles and promotes the conversion of HDL₃ to HDL₂. The observation that HDL₃ is preserved in plasma from nephrotic patients at the apparent expense of HDL₂ suggests that the LCAT activity is reduced in nephrotic syndrome.³⁹⁴ Furthermore, mature HDL also transports a number of apolipoproteins that serve as cofactors. One of these apolipoproteins, Apo C-II, is an endogenous activator of LPL activity. Apo C-II is normally transported by HDL₂ to nascent VLDL and chylomicrons. Apo C-II may be lost in the urine of nephrotic patients, either as free protein or bound to HDL.³⁹⁵ Additionally, an inhibitor of Apo C-II, Apo C-III, is increased in nephrotic syndrome; with the resulting decreased Apo C-II to Apo C-III ratio, the activity of LPL is significantly decreased.

CLINICAL CONSEQUENCES OF HYPERLIPIDEMIA

The most important consequence of hyperlipidemia is its potential for inducing cardiovascular disease. The changes that occur in blood lipoprotein composition in nephrotic syndrome,³⁷⁵ reduced HDL₂ cholesterol, a relative increase in HDL₃ cholesterol, and the massive increase in total cholesterol, mostly found in the LDL, IDL, and VLDL fractions, are likely to increase the risk of atherosclerotic disease.

Given the natural history of atherosclerosis, one would predict that the patient with a protracted form of nephrotic syndrome has the highest risk of dying from premature cardiovascular disease.³⁹⁶ Accelerated atherosclerosis has been reported in patients with proteinuria and hyperlipidemia and in some studies has been associated with a strongly increased incidence of cardiovascular disease

and stroke.³⁹⁶ One study reported an 85-fold increase in the incidence of ischemic heart disease in nephrotic patients.³⁹⁷ In another retrospective analysis of 142 patients with proteinuria greater than 3.5 g/day, the relative risk of myocardial infarction was found to be 5.5 and the risk of cardiac death 2.8 compared with age- and sex-matched controls.³⁹⁶

Numerous studies have indicated a potential role for hyperlipidemia in the progression of CKD. It was proposed that filtered lipoproteins might accumulate in the mesangium and promote sclerosis.³⁹⁸ In animals, lipogenic diets have been shown to induce focal sclerosis, and the extent of glomerular damage correlates with the serum cholesterol.³⁹⁹ Free and esterified cholesterol was found in the glomeruli of nephrotic rats, and a close correlation was noted between plasma cholesterol levels and the number of sclerosing glomerular lesions.⁴⁰⁰ Whether hyperlipidemia also plays a role in the progression of CKD in the human nephrotic syndrome has yet to be determined.

Although there is no specific indication to treat the qualitative abnormalities that characterize the lipid disorder of nephrotic syndrome, if it is anticipated that the duration of hyperlipidemia will be prolonged, it is wise to initiate therapy. Treatment of nephrotic patients with ACE inhibitors⁴⁰¹ results in a decline in both proteinuria and blood lipid levels even if plasma albumin concentration does not increase. The decline in blood lipid levels includes a decrease in total cholesterol, lipoprotein(a), a decrease in VLDL and LDL cholesterol, and a decrease in the activities of CETP and LCAT.⁴⁰²

It is prudent to restrict dietary cholesterol and saturated lipids in patients with nephrotic syndrome. The long-term effects of dietary supplementation with fish oil (rich in omega-3-polyunsaturated fatty acids) are unknown, and it cannot be recommended as standard treatment except within the context of a controlled investigative trial. If reduction of proteinuria and dietary fat restriction do not effectively reduce hyperlipidemia, a variety of lipid-lowering drugs including the 3-hydroxy-3-methyl-glutaryl coenzyme A reductase inhibitors (statins), antioxidants, and fibric acid derivatives may be useful.³⁷³ Novel therapies including inhibitors of PCSK9, in the form of monoclonal antibodies targeting PCSK9³⁹¹ and lipid apheresis,⁴⁰³ are under investigation in patients with refractory nephrotic syndrome.

HYPERCOAGULABILITY

Urinary loss of some of the proteins involved in the coagulation cascade and the adaptive increased synthesis of others can induce a hypercoagulable state.⁴⁰⁴ Although arterial thrombosis has been reported, it is venous thrombosis that occurs with a particularly high incidence in nephrotic subjects.⁴⁰⁵

PATHOGENESIS OF HYPERCOAGULABILITY

In nephrotic syndrome, there are widespread alterations in synthesis, turnover, and urinary losses of proteins involved in coagulation and fibrinolysis. Numerous coagulation abnormalities occur in nephrotic syndrome (Fig. 29.13). Alterations in the concentrations of almost every coagulation factor including zymogens (factors II, V, VII, IX, X, XI,

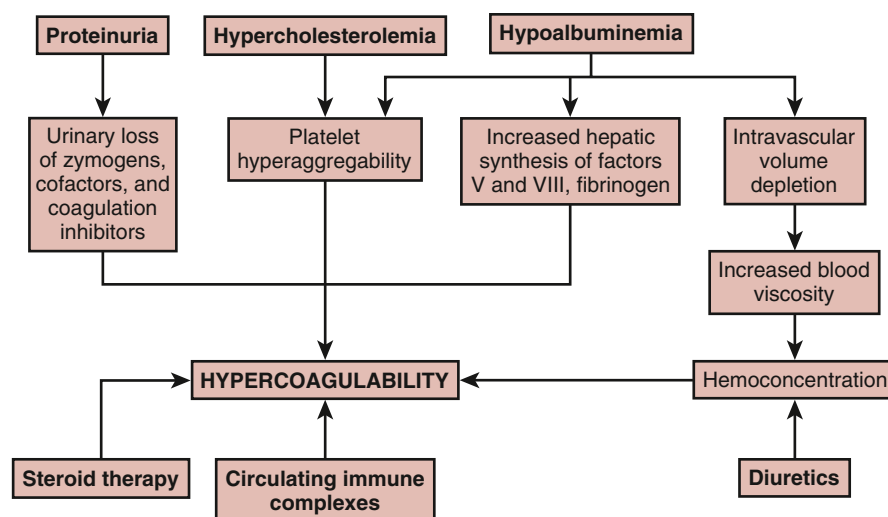


Fig. 29.13 Mechanisms in the pathophysiology of hypercoagulability in nephrotic syndrome. Altered levels and activity of factors in the intrinsic and extrinsic coagulation cascades, levels of antithrombotic and fibrinolytic components of plasma, platelet count and function, and other factors, such as steroids or diuretics, are the numerous abnormalities that contribute to hypercoagulability in nephrotic syndrome.

and XII), cofactors (factors V and VIII), and fibrinogen can occur.⁴⁰⁵ Plasma proteins lost in the urine in patients with nephrotic syndrome include factors IX, X, and XII, which become deficient as there is no sufficient increase in synthetic rates.⁴⁰⁵ In contrast, proteins of higher molecular weight, including factors V, factor VIII, and fibrinogen accumulate because of increased synthesis.⁴⁰⁴ Levels of factor VIII typically increase as much as twofold to threefold.⁴⁰⁴ However, because factor VIII is also an acute phase reactant, high factor VIII levels may be an epiphenomenon rather than a causal factor in the development of venous thrombosis. Furthermore, in nephrotic syndrome, microparticles released from cell membrane with phosphatidyl serine exposure of red blood cells, platelets, and endothelial cells provide binding sites for several activated clotting factors including factor Xa, contributing at least partly to the thrombophilic susceptibility of these patients.⁴⁰⁶

There is an inverse correlation between serum albumin and fibrinogen levels in nephrotic syndrome.⁴⁰⁷ The elevated plasma levels of fibrinogen likely result from increased hepatic synthesis, as catabolism is normal.⁴⁰⁸ Hyperfibrinogenemia may contribute to the procoagulant state by providing more substrate for fibrin formation and by promoting platelet hyperaggregability, increased blood viscosity, and red blood cell aggregation. Increased fibrin deposition, however, may also occur due to increased thrombin formation by the elevated levels of factors V and VIII.⁴⁰⁹

Nephrotic patients exhibit abnormalities in endogenous coagulation inhibitors including antithrombin III, which is deficient in 40% to 80% of patients.⁴¹⁰ Alterations in other endogenous anticoagulants such as protein S, protein C, and tissue factor pathway inhibitor (TFPI) may also occur in patients with nephrotic syndrome, but the findings are conflicting.⁴¹¹

Several factors may lead to a reduction in plasmin-induced fibrinolysis in nephrotic syndrome; much of the work has focused on plasminogen, the precursor for plasmin, and two major regulators of plasmin formation, plasminogen activator inhibitor (PAI-1) and tissue plasminogen activator (t-PA).

Maintenance of hemostasis also involves the formation of platelet plugs through platelet activation and aggregation. Studies examining platelet abnormalities have suggested a role for enhanced, platelet-vessel wall interaction and platelet aggregation in the development of thromboembolism in nephrotic syndrome. Thrombocytosis, decreased red blood cell deformability, and increased von Willebrand factor levels all favor platelet transport toward the vessel wall and increased platelet adhesion⁴¹² and are observed in nephrotic syndrome.

Platelet hyperaggregability is associated with hypoalbuminemia, hypercholesterolemia, and hyperfibrinogenemia.⁴¹³

To date, observations suggest that platelet activation and aggregation may play a role in the increased risk of thromboembolism in patients with nephrotic syndrome. However, attempts at correlating in vitro functional tests with clinically overt thromboembolic events have shown conflicting results.⁴¹⁴

Other clinical features of the nephrotic state, such as intravascular volume depletion and exposure to steroids, also contribute to hypercoagulability. Increased blood viscosity is associated with hemoconcentration and enhanced by the use of diuretics⁴¹⁵ and by hyperfibrinogenemia.⁴⁰⁴ The nature of the underlying immunologic injury may play a role and account for the predilection of thrombosis for the renal vein and increased incidence of thrombotic complications in membranous glomerulopathy.⁴⁰⁵ The use of steroids has also been suggested to predispose patients to thromboembolic complications,⁴⁰⁵ but other studies have reported a high incidence of thromboembolic complications in the absence of steroid therapy as well.⁴¹⁶

In addition to acquired risk factors, the process of thromboembolism in nephrotic syndrome may also involve undetermined genetic factors. A recent retrospective observational study in 36 patients with nephrotic syndrome reported the prevalence of factor V Leiden G1691A polymorphism, prothrombin G20210A mutation, the 4G/5G mutation of plasminogen activator inhibitor (PAI) gene, and each of the two methylene tetrahydrofolate reductase (MTHFR) gene polymorphisms (C677T and

A1298C) being 14%, 5.6%, 44.5%, and 27.8%, respectively, higher than that reported in the general population.⁴¹⁷ On multivariate analysis, the presence of at least two mutations was independently associated with the risk of venous thromboembolic events.

Thus abnormalities in any of the steps that promote coagulation including activation and termination of the coagulation cascade, fibrinolysis, and platelet activation and aggregation may contribute to the hypercoagulable state seen in nephrotic syndrome. The specific role of each of these alterations remains ill-defined.

CLINICAL CONSEQUENCES OF HYPERCOAGULABILITY

Thromboembolic events are serious complications of nephrotic syndrome. The most frequent site of thrombosis is the renal vein. Retrospective and prospective studies have shown an incidence of renal vein thrombosis (RVT) in nephrotic syndrome ranging from 5% to 62%.⁴⁰⁵ Nephrotic syndrome is associated with RVT regardless of the underlying disease. Observational studies evaluated patients who underwent renal venography.^{418,419} These studies show that the prevalence of RVT is highest in patients with membranous nephropathy, on average 37%. However, the risk is still clinically important in other primary glomerular diseases, particularly in membranoproliferative glomerulonephritis and minimal-change disease. Furthermore, the risk of developing RVT may have been underestimated in these largely cross-sectional studies, since patients who were initially not found to have RVT may have subsequently developed one during the disease. In the largest prospective study assessing RVT in 151 patients with nephrotic syndrome, the cumulative incidence of RVT was 22%.⁴¹⁸

RVT presents clinically in two ways,⁴¹⁸ acute and chronic. Acute RVT is usually unilateral and characterized by acute flank pain, flank tenderness, macroscopic hematuria, and some deterioration of renal function. Chronic RVT is usually asymptomatic and occurs in the elderly. Selective renal venography is the gold standard for the diagnosis of RVT, and demonstration of venous collaterals establishes chronicity. However, renal venography is invasive and associated with complications that include pulmonary embolism due to clot dislodgement, inferior vena cava perforation, and contrast-induced acute kidney injury.⁴²⁰ Consequently, noninvasive diagnostic tests are preferred including intravenous pyelography, computed tomography, and magnetic resonance imaging.⁴²⁰ Nevertheless, further studies are necessary because the usefulness of these techniques in the diagnosis or exclusion of acute RVT remains unproven. Doppler ultrasonography appears to be inferior to renal venography in establishing the diagnosis of RVT and cannot be recommended on the basis of current data.⁴²¹

Deep venous thrombosis of the lower extremities is also observed in nephrotic syndrome and can occur in isolation in up to 15% of patients⁴¹⁹ or in association with RVT.⁴¹⁸

Pulmonary embolism may complicate deep vein thrombosis involving the lower extremities, inferior vena cava, or RVT. In a case series of 151 patients with the nephrotic syndrome,⁴¹⁸ of whom 94 patients underwent ventilation-perfusion lung scanning, symptomatic pulmonary embolism was observed in 25% of patients with acute RVT and 20% of

patients with chronic RVT. The incidence of asymptomatic pulmonary embolism was 12.8%. In both prospective and retrospective studies, the incidence of thromboembolic complications other than RVT ranges from 8% to 44%, with an average incidence of 20%.⁴⁰⁵

The pattern of thrombosis is remarkably different between children and adults. Despite a lower incidence (1.8% to 5%),⁴²² thromboembolic complications in children tend to be more severe, and half of the children have arterial thrombosis that may cause clinical problems, such as persistent hemiplegia, mesenteric infarction, and peripheral occlusion leading to amputation.⁴²² In adults, arterial thrombosis is much less common than venous thrombosis, but it is a serious complication causing important morbidity. One case series described 43 patients with the nephrotic syndrome who had arterial thromboembolism at aortic, renal, femoral, mesenteric, cerebral, or brachial sites.⁴¹⁶ An increased risk of coronary events in patients with the nephrotic syndrome has been documented in a retrospective study.³⁹⁶

The treatment of venous thromboembolism in nephrotic patients is like that in the general population. First-line treatment consists of conventional anticoagulation with low-molecular-weight heparin and oral vitamin K antagonists.⁴⁰⁵ When therapy is initiated early in the course of acute RVT, renal function and other symptoms of RVT have been shown to improve significantly.⁴¹⁸ Oral vitamin K antagonists are usually continued for the duration of nephrotic-range proteinuria⁴²³ because RVT can recur in the setting of ongoing nephrosis after withdrawal of anticoagulation therapy. Low-molecular-weight heparins have a reasonable safety profile but should be used cautiously in patients with renal insufficiency because of excessive anticoagulant activity and an increased risk of bleeding due to drug accumulation.^{424,425}

Currently, direct oral anticoagulants (DOACs) are not widely used in nephrotic syndrome. These agents are direct factor Xa inhibitors (rivaroxaban, apixaban, edoxaban, betrixaban) or direct factor IIa (prothrombin) inhibitors, such as dabigatran, and have been recommended for the management of acute deep vein thrombosis.⁴²⁶ However, in patients with nephrotic syndrome, many uncertainties with the use of DOACs still remain, such as dose and monitoring and changes in pharmacokinetics and pharmacodynamics related to hypoalbuminemia, which need further studies.⁴²⁷

Controversy exists regarding the use of prophylactic anticoagulation therapy in patients with nephrotic syndrome who do not have RVT.⁴⁰⁷ Prophylactic anticoagulation is warranted as long as the patient has nephrotic proteinuria, an albumin level below 2 g/dL, or both. In patients with other underlying diseases, a more cautious approach may be indicated, and prophylactic anticoagulant therapy should be initiated only if the risk of thromboembolic events is considered high.

SUSCEPTIBILITY TO INFECTION

Loss of high filtered protein load through both urinary excretion and tubular catabolism,⁴²⁸ as well as a reduced rate of synthesis,⁴²⁹ may result in concurrent deficiency of IgG and components of the alternate complement pathway,

including factor B. Indeed, patients with nephrotic syndrome have low serum levels of various IgG subclasses. Also, IgA levels are decreased in nephrotic syndrome, whereas IgM usually is increased, particularly in patients with minimal-change disease and normal renal function.⁴³⁰ Furthermore, defective cell-mediated immunity has been reported in nephrotic syndrome^{431,432} including reduced number of total circulating T lymphocytes and blunted blastogenic response by lymphocytes to the mitogens concanavalin A and phytohemagglutinin.

The defects in humoral and cellular-mediated immunity render the nephrotic patient highly susceptible to infection.⁴³³ The organisms most frequently encountered are *Streptococcus pneumoniae* and *Escherichia coli*. Although such susceptibility to infection is generalized, there seems to be a particular vulnerability to local infection at the sites of edema formation. Splits in the skin caused by edema and malnutrition may predispose nephrotic patients to cellulitis.³⁴⁰ Peritonitis has been reported in patients who have ascites.⁴³³ It also occurred in approximately 6% of children with nephrotic syndrome who suffered one or more episodes of the infection.⁴³³

The unusual susceptibility of children to infections with encapsulated microorganisms is associated with urinary losses of the alternate pathway complement components, particularly factor B, or C3 proactivator, and D, which are essential for the destruction of encapsulated bacteria in the absence of specific antibody.⁴²⁸ The capacity to opsonize the encapsulated bacteria can be restored to normal by addition of pure factor B to nephrotic serum.⁴²⁸

New potent antibiotics have contributed to considerably decrease the incidence of fatal infections in nephrotic syndrome. However, prophylactic measures, such as pneumococcal vaccine, are recommended in adults with severely depressed immunoglobulin levels and nephrotic children older than 2 years of age, especially when early remission of nephrotic syndrome is not anticipated.⁴³⁴

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QUESTIONS

- Charge selectivity in the glomerular basement membrane (GBM) functions in the following manner:
 - Positive charges on the GBM repel positively charged circulating proteins
 - Positive charges on the GBM attract negatively charged circulating proteins
 - Negative charges on the GBM attract positively charged circulating proteins
 - Negative charges on the GBM repel negatively charged circulating proteins

Answer: d

Rationale: The majority of circulating proteins including albumin are negatively charged. Evidence from theoretical models and experiments in animals suggests that negatively charged molecules in the glomerular capillary wall (in the glycocalyx of endothelial cells, the negatively charged heparansulfate of the glomerular basement membrane and the glycoproteins of the cell membrane of podocytes) form an important component of glomerular permselective function.

- The components of the glomerular capillary membrane wall are:
 - Endothelial cells and glomerular basement membrane
 - Endothelial cells and basement membrane and podocyte foot process
 - Endothelial cells and podocyte foot process
 - Endothelial cells and basement membrane and parietal cells
 - Endothelial cells and parietal cells and podocyte foot processes

Answer: b

Rationale: The glomerular capillary membrane is a complex structure composed of the three layers stated in option (b) that allows a large amount of water and small solute filtration while efficiently restricting glomerular passage of protein macromolecules within blood circulation.

- The following molecules are thought to be important in the absorption of filtered proteins by proximal tubule cells: (select all that apply)
 - Uromodulin
 - Megalin
 - C3 complement
 - Clathrin
 - Nephrin

Answer: b and d

Rationale: Renal proximal tubule cells have a remarkable ability to reabsorb large quantities of albumin through clathrin- and megalin receptor-mediated endocytosis. Uromodulin (Tamm-Horsfall protein) is secreted by tubule cells and is thought to play a role in preventing urinary infections. C3 is a component of the complement system that is believed to play an important role in the progressive renal injury resulting from proteinuria. Nephrin is a key component of the glomerular slit diaphragm.

- Locally produced angiotensin II contributes to glomerular injury by:
 - Sustaining glomerular hypertension-induced damage
 - Impairing the glomerular barrier-sieving function through inhibition of nephrin expression by podocytes
 - Activating parietal epithelial progenitor cells in the setting of glomerular hyperplastic lesions
 - All the above mechanisms

Answer: d

Rationale: Angiotensin II has been identified as a key mediator of multiple mechanisms that contribute to proteinuria and progressive kidney damage including all those listed.

- The main systemic consequences of nephrotic-range proteinuria include:
 - Alteration in composition of the body protein pool—hypoalbuminemia
 - A state of sodium retention
 - Dyslipidemia
 - Abnormalities of coagulation factors
 - Susceptibility to infection
 - All the above abnormalities

Answer: f

Rationale: The nephrotic syndrome is a complex pathophysiologic state characterized by sodium and water retention, as well as multiple abnormalities resulting from excessive urinary loss of albumin and other proteins that are components of the coagulation and humoral immune systems.